Chapter 2: The Physical Layer

1 What is the basic goal of the physical layer?

2 How to define the physical layer protocols?

3 What is the theoretical basis for the physical layer?

*4 What are the commonly used transmission media?

*5 What are the common physical layer interface standards?

第二章：物理层

1 物理层的基本目标是什么？

2 如何定义物理层协议？

3 物理层的理论基础是什么？

*4 常用的传输介质有哪些？

*5 常见的物理层接口标准有哪些？

Keypoints and Difficulties

Keypoints:

- Nyquist's Law and Shannon-Hartley theorem
- Symbol code types: NRZ, Manchester encoding, Differential Manchester encoding

Difficulties:

- Channel transmission rate calculation

要点与困难

要点：

- 奈奎斯特定律与香农-哈特利定理
- 符号代码类型：NRZ，曼彻斯特编码，差分曼彻斯特编码

困难：

- 信道传输速率计算

2-2

1 The Goals

- To ensure the correctness of the transmitted signals "0" and "1" and the consistency of transmission and reception;
- Bit – transmission mode, speed, duration, and signal distortion;
- Interface design: the number of pins, specifications, functions, etc.;
- Signal transmission procedures: how to arrange the transmission process and the order of events;

1 目标

- 为确保传输的 "0" 和 "1" 信号的正确性以及传输与接收的一致性；
- 比特 – 传输模式、速度、持续时间和信号失真；
- 接口设计：引脚数量、规格、功能等；
- 信号传输流程：如何安排传输过程和事件顺序；

Four Important Characteristics

- Mechanical Characteristics
  - Specifies the size of the connector used in the physical connection, the number and arrangement of pins.
- Electrical Characteristics
  - Specifies the transmission mode, voltage level, coding, impedance matching, transmission rate and distance limit when transmitting the binary bit.

四个重要特性

- 机械特性
  - 规定了物理连接中使用的连接器尺寸、引脚数量和排列方式。
- 电气特性
  - 指定传输二进制位时的传输模式、电压电平、编码、阻抗匹配、传输速率和距离限制。

Four Important Characteristics

- Functional Characteristics
  - Define the function of each physical line, indicating the means of a special voltage appears on a line
  - Line functions are divided into four categories: Data, Control, Timing, Power Supply
- Procedural Characteristics
  - Define the working rules and timing relationships for each physical line
  - Signal transmission: simplex, half duplex, full duplex

四个重要特性

- 功能特性
  - 定义每条物理线的功能，标明特殊电压在线路上的出现方式。
  - 线功能分为四类：数据、控制、时序、电源
- 过程特性
  - 定义每条物理线的工作规则与时序关系
  - 信号传输：单工、半双工、全双工



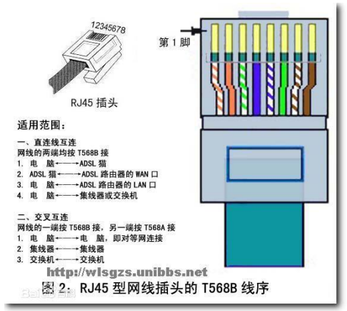


图 2：RJ45 型网线插头的 T568B 线序



1 TX+（数据发送正端）
2 TX-（数据发送负端）
3 RX+（数据接收正端）
4 未用
5 未用
6 RX-（数据接收负端）
7 未用
8 未用



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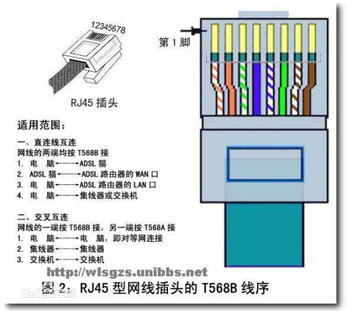


图 2：RJ45 型网线插头的 T568B 线序



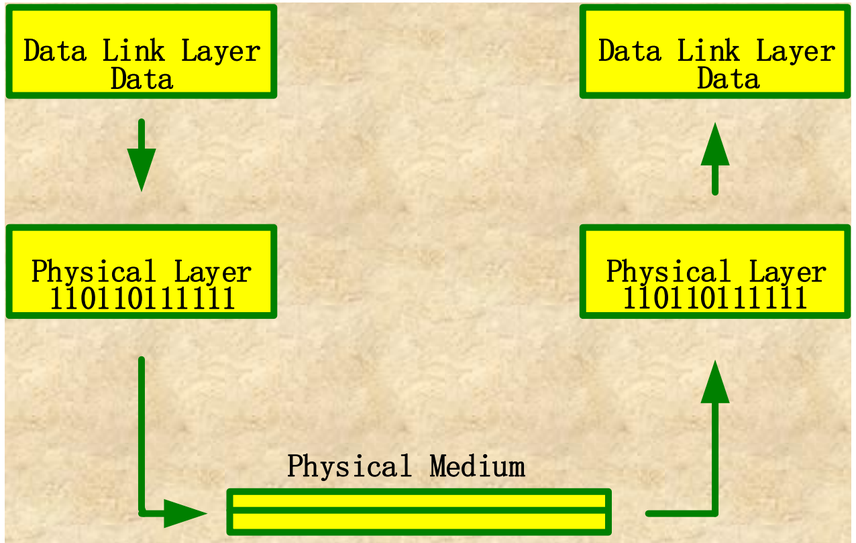
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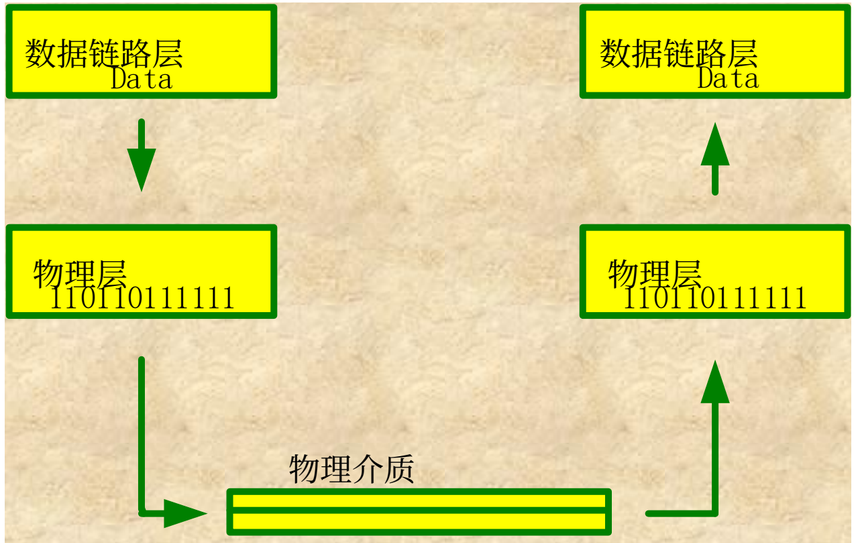
2 Contents of Physical Layer Protocol

- Responsible for reliably transmitting bit data from one end of the physical medium to the other



2 物理层协议内容

- 负责在物理介质的两端之间可靠地传输比特数据



3 Basic theory of data communication

(1) Basic concepts

(2) For a specific physical communication channel, what is the maximum transmission rate? Infinite?

(3) How to transmit Bits in specific physical communication channels? Transmission speed, transmission duration, signal distortion?

(4) In order to save communication equipment and costs, how to transmit the information of multiple computers in one physical channel?

3 数据通信基本理论

(1) 基本概念

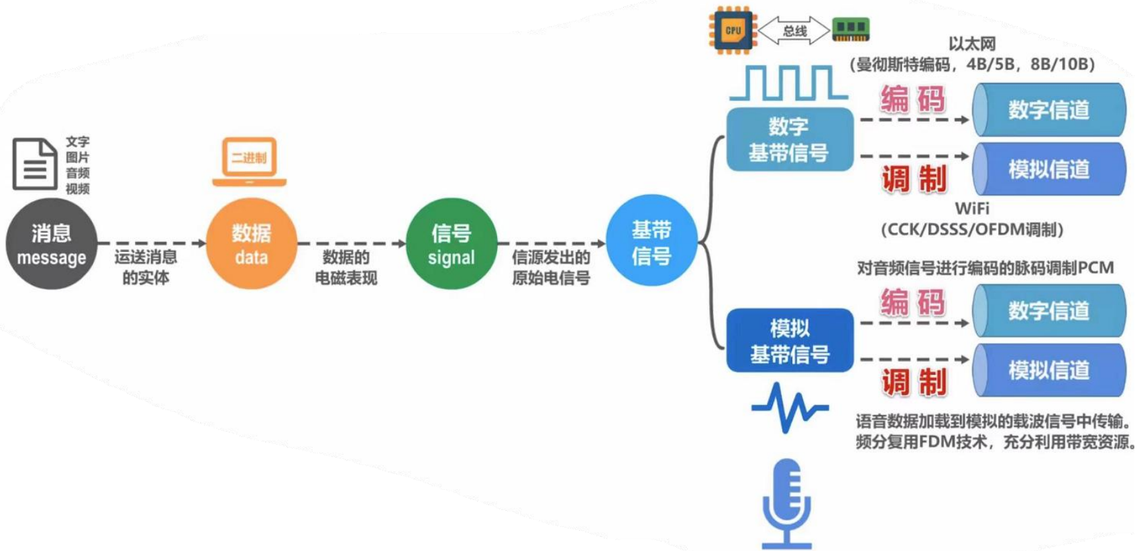
(2) 对于特定的物理通信信道，最大传输速率是多少？无限？

(3) 如何在特定的物理通信信道中传输比特？传输速率、传输时延、信号失真？

(4) 为了节省通信设备和成本，如何在一条物理信道中传输多台计算机的信息？

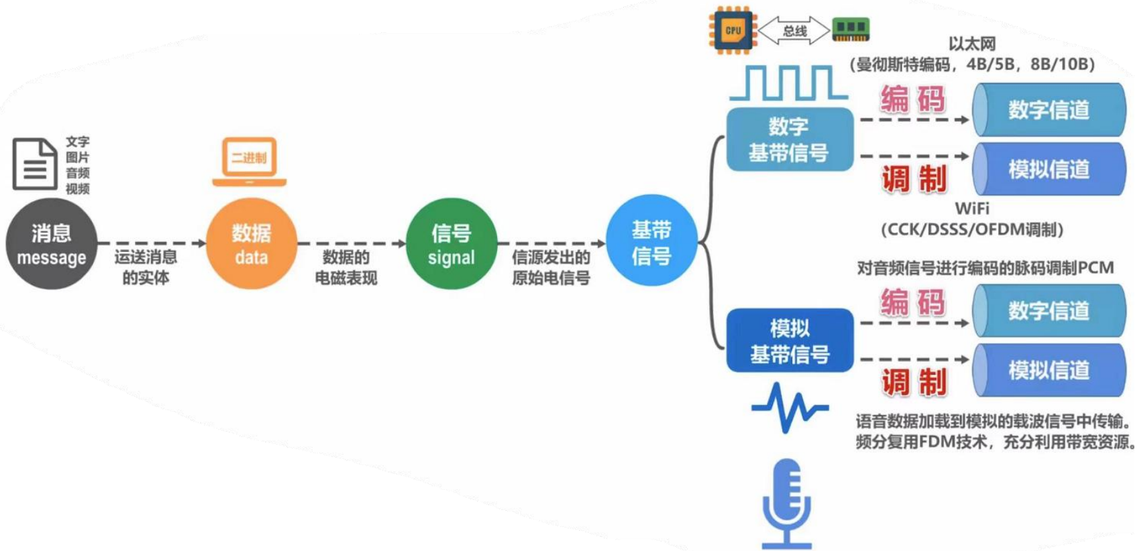
(1) Basic concepts

- Data: Physical symbols describing objective facts
- Signal: Form of data transmission process
- Channel: media that sends information in a certain direction



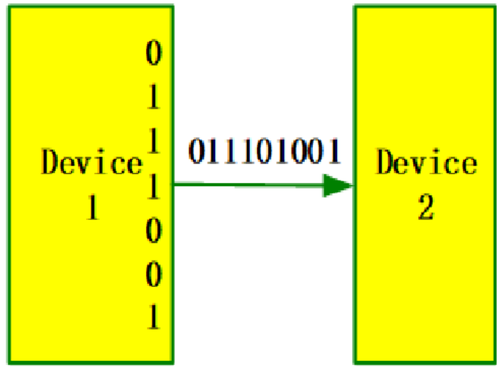
(1)基本概念

- 数据：描述客观事实的物理符号
- 信号：数据传输过程的形式
- 信道: 媒体，用于向特定方向发送信息



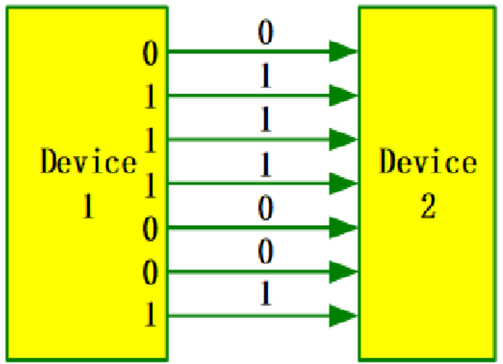
(1) Basic concepts

Serial transmission



- Suitable for long distance communication

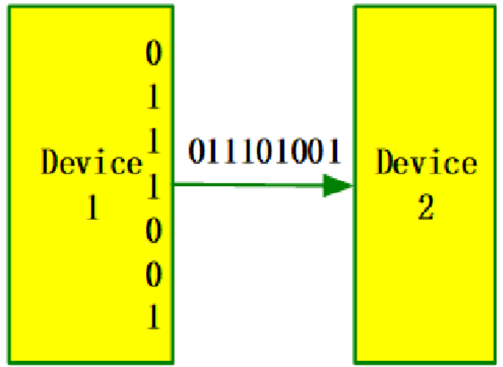
Parallel transmission



- Suitable for short distance communication

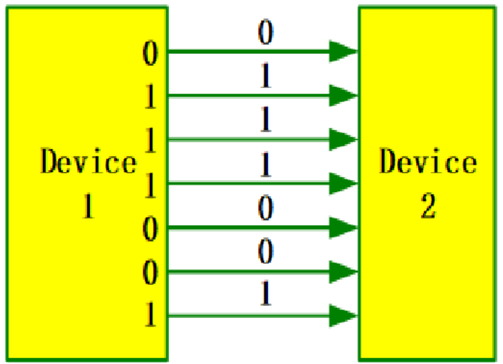
(1) 基本概念

串行传输



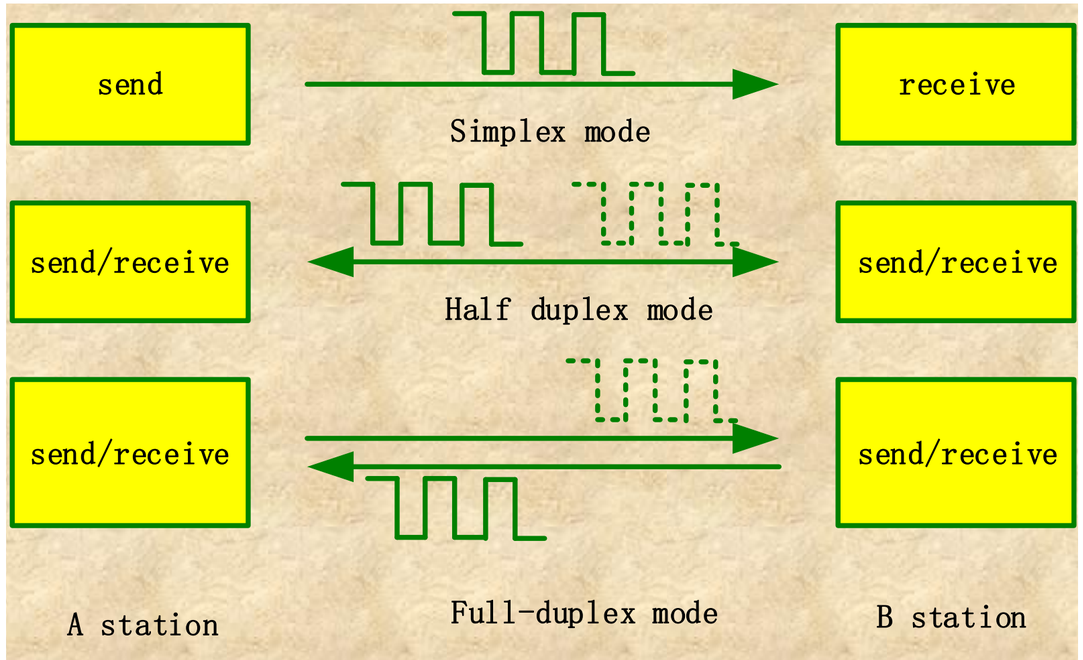
- 适合远距离通信

并行传输

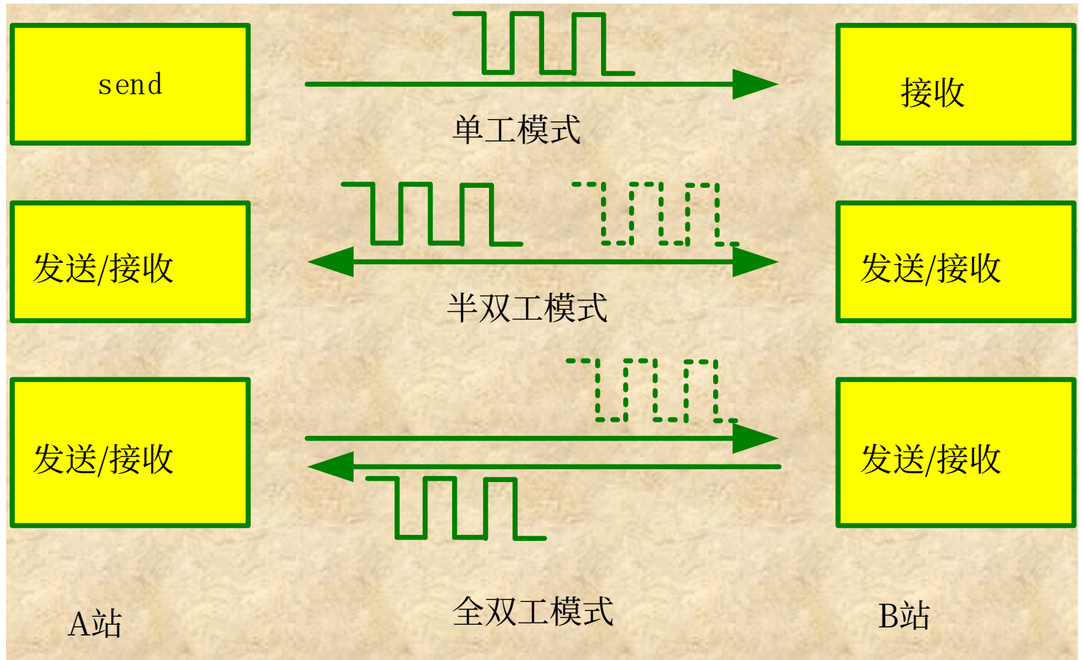


- 适合短距离通信

(1) Basic concepts



(1) 基本概念



3 Basic theory of data communication

(1) Basic concepts

(2) For a specific physical communication channel, what is the maximum transmission rate? Infinite?

(3) How to transmit Bits in specific physical communication channels? Transmission speed, transmission duration, signal distortion?

(4) In order to save communication equipment and costs, how to transmit the information of multiple computers in one physical channel?

3 数据通信基本理论

(1) 基本概念

(2) 对于特定的物理通信信道，最大传输速率是多少？无限吗？

(3) 如何在特定的物理通信信道中传输比特？传输速度、传输持续时间、信号失真？

(4) 为了节省通信设备和成本，如何在一条物理信道中传输多台计算机的信息？

(2) Channel Characteristics

- Symbol: A digital pulse, which represents the basic waveform of a discrete value
  - ASCII code of letter 'A': 1000001, can be represented by 7 pulses/symbols
- The amount of information carried by a symbol is determined by the number of discrete values taken by the symbol
  - Two discrete values ("0" or "1"): 1 bit
  - Four discrete values for a symbol: 2 bits
  - If N discrete values for a symbol of n (bits), then

$$n = \log_2 N$$

(2) 信道特性

- 符号：一种数字脉冲，代表离散值的基本波形
  - 字母 'A' 的ASCII代码：1000001，可以表示为7个脉冲/符号
- 一个符号携带的信息量由该符号取的离散值数量决定
  - 两个离散值（"0"或"1"）：1比特
  - 一个符号有四个离散值：2比特
  - 如果符号n（比特）有N个离散值，那么

$$n = \log_2 N$$

(2) Channel Characteristics

- Baud rate (modulation rate or symbol rate):
  - The number of times the signal changes every second
    - 1 baud sends one symbol per second
- Bit rate:
  - The number of bits transmitted per second
- Relationship between them:
  - If each symbol has n bits: bit rate (b/s) = baud rate * n
  - If each symbol has 3 bits, then the bit rate is three times the baud rate
  - If each symbol has 1 bit, the bit rate and baud rate are the same

(2) 信道特性

- 波特率（调制速率或符号速率）：
  - 信号每秒变化的次数
    - 1 波特每秒发送一个符号
- 波特率：
  - 每秒传输的比特数
- 它们之间的关系：
  - 如果每个符号有 n 位： 比特率 (b/s) = 波特率 * n
  - 如果每个符号有 3 位，那么比特率是波特率的 3 倍
  - 如果每个符号有1位，那么比特率和波特率相同

(2) Channel Characteristics

- Channel capacity (bandwidth in networks)
  - The maximum number of bits that can be transmitted over a channel per unit time, expressed in bps
- Frequency bandwidth (in communication)
  - Frequency range of the signal allowed by the channel (= maximum frequency - minimum frequency), in Hz
  - E.g., people can hear sound waves of 20Hz ~ 20kHz, i.e., frequency bandwidth of hearing system is19980Hz

(2) 信道特性

- 信道容量（网络中的带宽）
  - 信道每单位时间内能传输的最大比特数，以bps表示
- 频率带宽（通信中）
  - 信道允许的信号频率范围(= 最大频率 - 最小频率), 单位为Hz
  - 例如，人们可以听到20Hz ~ 20kHz的声波，即听觉系统的频率带宽是19980Hz

(2) Channel Characteristics

- Nyquist's Law :

  1) For an ideal low-pass channel with the frequency bandwidth of W (Hz), its maximum symbol transmission rate = 2W Baud

  2) For an ideal band-pass channel with the frequency bandwidth of W (Hz), its maximum symbol transmission rate = W Baud

- Question
  - The frequency band of a standard telephone channel is 300 ~ 3400Hz, i.e., the frequency bandwidth is 3100Hz. What is the maximum symbol transmission rate through the ideal low-pass channel?

(2) 信道特性

- 奈奎斯特定律：

  1) 对于频率带宽为 W (Hz) 的理想低通信道，其最大符号传输速率为 = 2W Baud

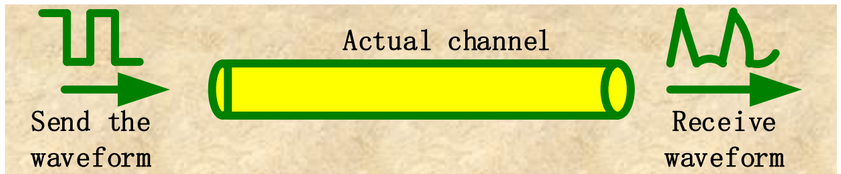
  2) 对于频率带宽为 W (Hz) 的理想带通信道，其最大符号传输速率为 = W Baud

- 问题
  - 标准电话信道的频率带宽是300 ~ 3400Hz，即频率带宽为3100Hz。理想低通信道的最大符号传输速率是多少?

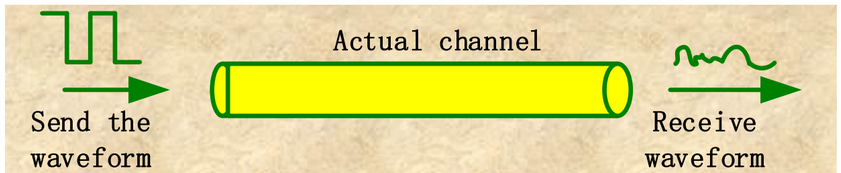
(2) Channel Characteristics

- Actual channels are not ideal – Distortion exists
  - Higher symbol transmission rate
  - Farther signal transmission distance

  } Larger distortion
- Small and recognizable distortion



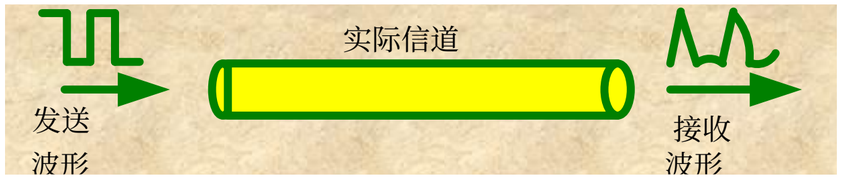
- Large and unrecognizable distortion



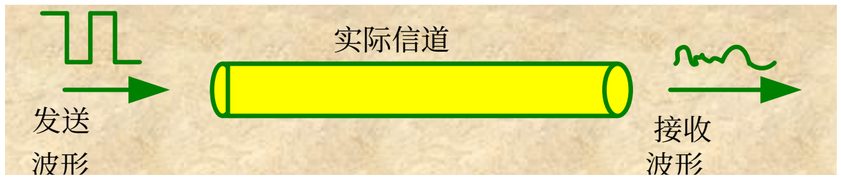
(2) 信道特性

- 实际信道并非理想——存在失真
  - 更高的符号传输率
  - 更远的信号传输距离

  } 更大的失真
- 小且可识别的失真



- 大且不可识别的失真



(2) Channel Characteristics

- Nyquist's Law :

  1) For an ideal low-pass channel with the frequency bandwidth of W (Hz), its maximum symbol transmission rate = 2W Baud

  2) For an ideal band-pass channel with the frequency bandwidth of W (Hz), its maximum symbol transmission rate = W Baud

- Question
  - The frequency band of a standard telephone channel is 300 ~ 3400Hz, i.e., the frequency bandwidth is 3100Hz. What is the maximum symbol transmission rate through the ideal low-pass channel?

(2) 信道特性

- 奈奎斯特定律：

  1) 对于一个理想的 低通信道 其频率 带宽为 W (Hz)，其最大符号传输速率 = 2W 波特

  2) 对于一个理想的 带通信道 其频率 带宽为 W (Hz)，其最大符号传输速率 = W 波特

- 问题
  - 标准电话信道的频率带宽是300 ~ 3400Hz，即频率带宽为3100Hz。理想低通信道的最大符号传输速率是多少?

(2) Channel Characteristics

- Shannon's Formula: limit information transmission rate C of channel can be expressed as:

$$C = W\log_2(1+S/N)$$

  - W - Frequency bandwidth in Hz
  - S - Average signal power through the channel
  - N - Gaussian noise power through the channel
  - S/N – related to signal-to-noise ratio
    - Typically in db (decibels): $10\log_{10}S/N$
- Actual rate much lower than C due to signal loss
- Possible to achieve error-free transmission as long as transmission rate < C

(2) 信道特性

- 香农公式：信道的信息传输速率C的上限可以表示为：

$$C = W\log_2(1+S/N)$$

  - W - 频率带宽（Hz）
  - S - 通过信道的平均信号功率
  - N - 通过信道的高斯噪声功率
  - S/N – 与信噪比相关
    - 通常为 分贝(分贝): $10\log_{10}S/N$
- 实际速率因信号损失远低于 C
- 只要传输速率< C，就有可能实现无误差传输

(2) Channel Characteristics

- Example:
  - For the standard telephone channel with frequency bandwidth of 3.1 kHz, if the signal-to-noise ratio S/N = 2500, then can the information transmission rate be 50kb/s?
- Solution:
  - Put W = 3.1kHz, S / N = 2500 into Shannon's Formula
  - The limit information transmission rate is 35kb/s, so it is impossible to reach 50kb/s
- In order to achieve 50kb/s, you can
  - improve the signal-to-noise ratio in the channel
  - or increase the frequency bandwidth

(2) 信道特性

- 示例：
  - 对于频率带宽为

3 1 = kHz的标准电

2500

话信道，如果信噪比S/N，那么信息传输速率能否达到50kb/s?

- 解：
  - 将 W = 3.1kHz、S / N = 2500 代入香农公式
  - 极限信息传输速率是 35kb/s，因此无法达到 50kb/s
- 要实现 50kb/s，你可以
  - 提高信道中的信噪比
  - 或者增加频率带宽

3 Basic theory of data communication

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3 数据通信基本理论

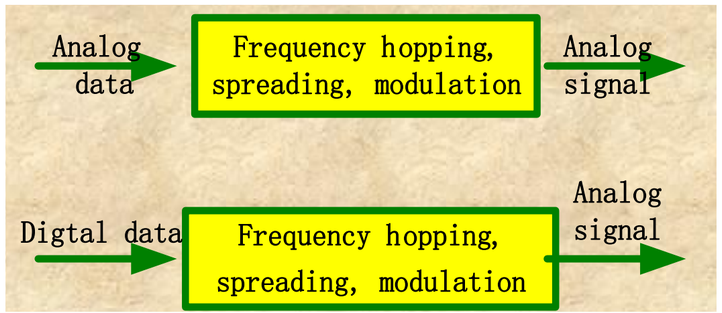
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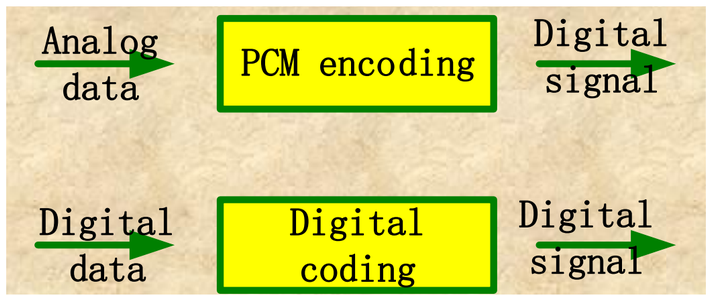
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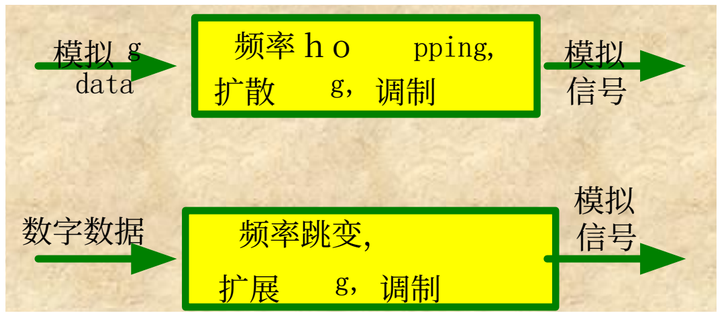
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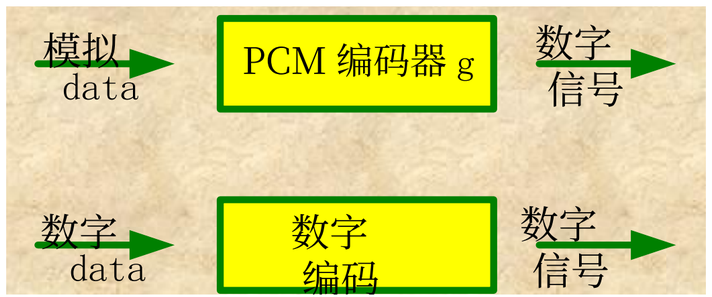
Techniques in analog/digital transmission



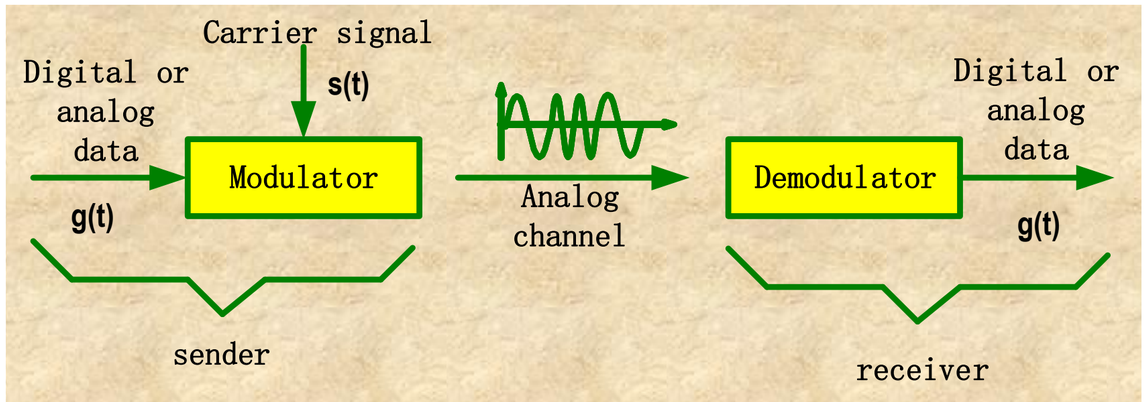


模拟/数字传输技术

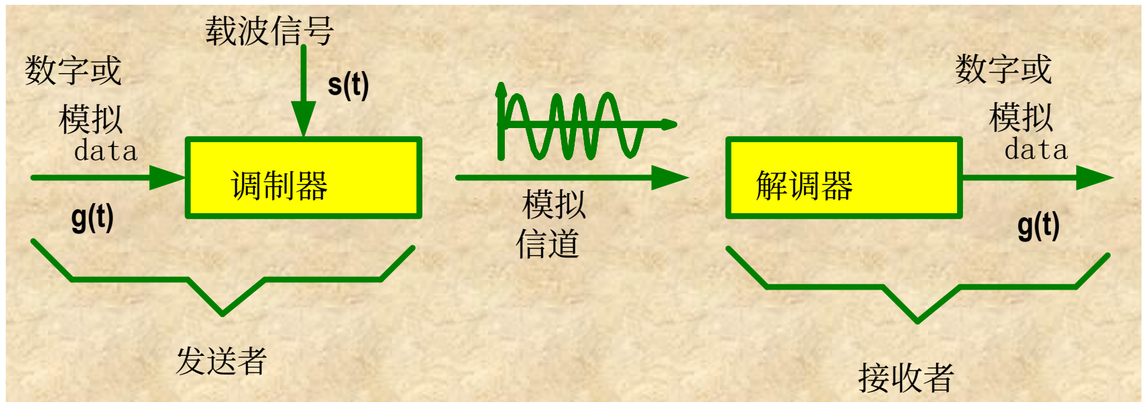




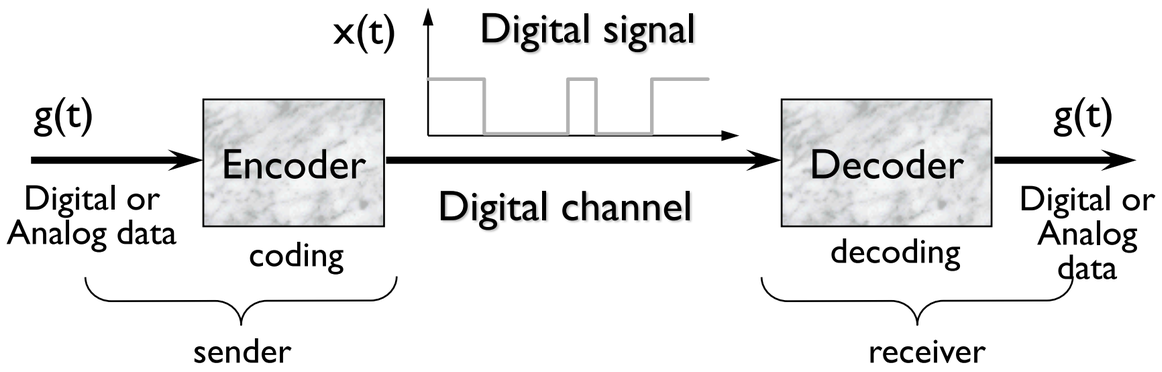
Modulation / demodulation system model



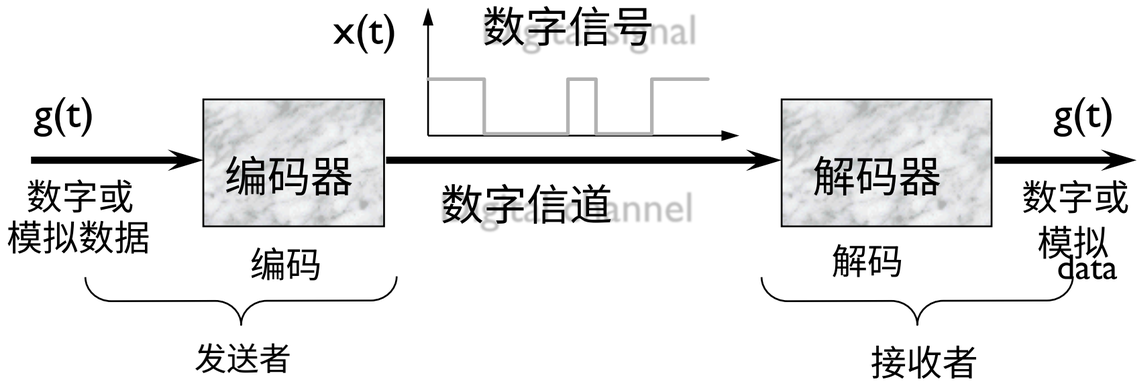
调制/解调系统模型



Coding / decoding system model

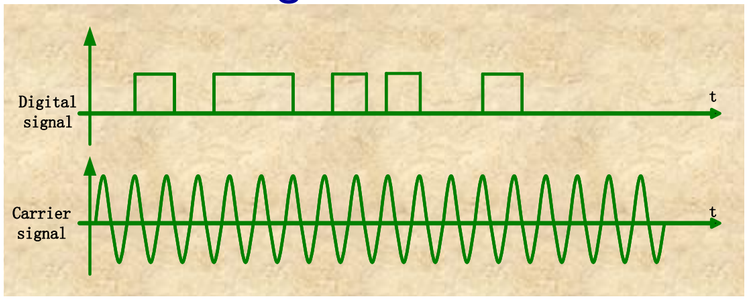


编码/解码系统模型

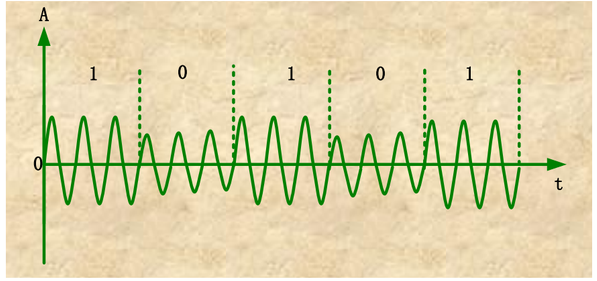


Modulation
(digital data to analog signals)

- Digital and carrier signals

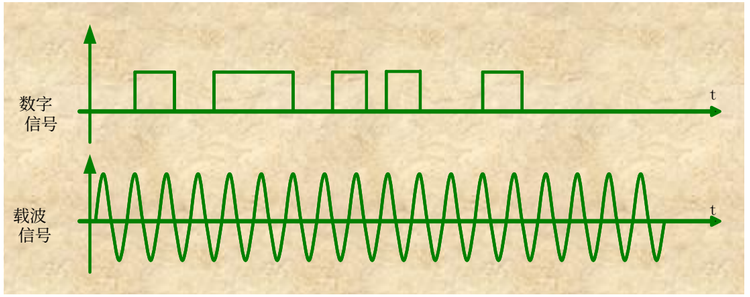


- Amplitude Modulation (AM)

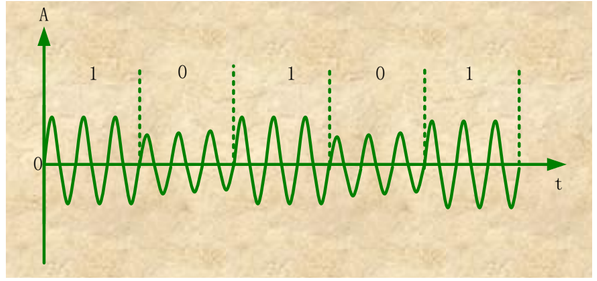


调制（数字数据到模拟信号）

- 数字与载波信号

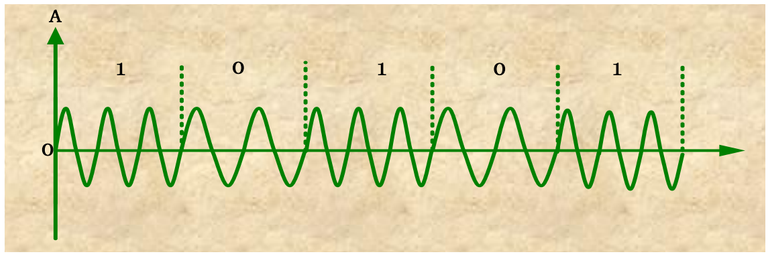


- 幅度调制 (AM)

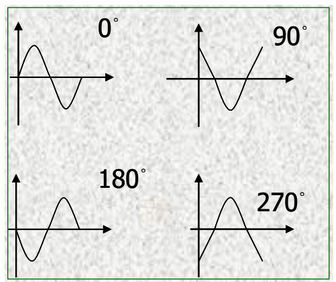


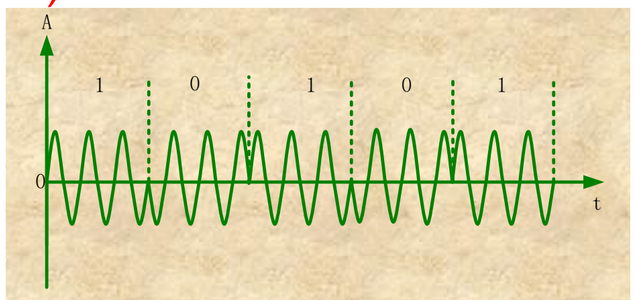
Modulation
(digital data to analog signals)

- Frequency Modulation (FM)



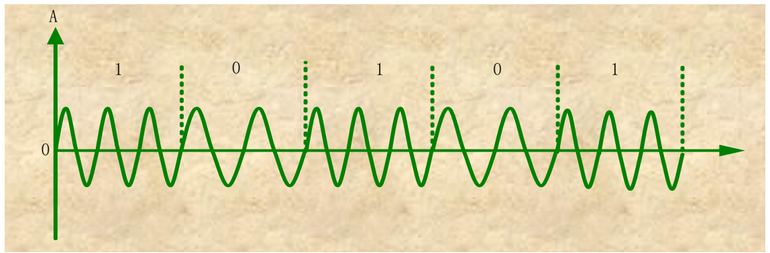
- Phase Modulation (PM)



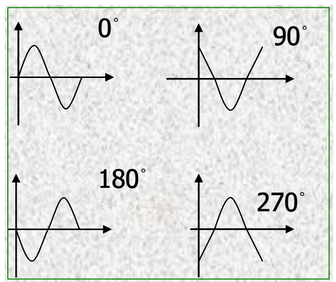


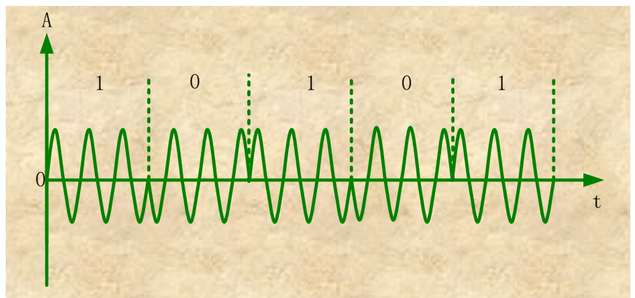
调制(数字数据到模拟信号)

- 频率调制 (FM)



- 相位调制 (PM)

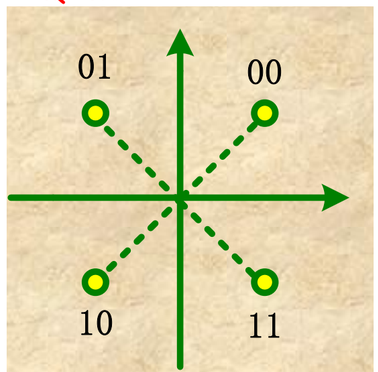




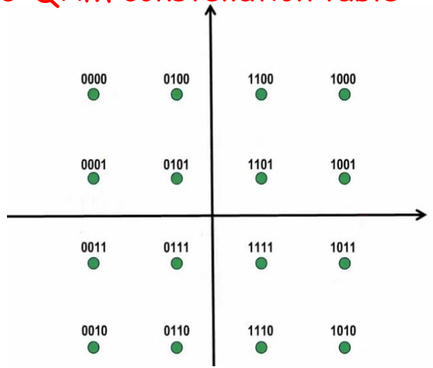
Modulation
(digital data to analog signals)

- Quadrature Amplitude Modulation (QAM)
  - PM and AM techniques
  - If there are x changes in phase and y changes in amplitude, then there are x * y combined changes

4-QAM constellation table



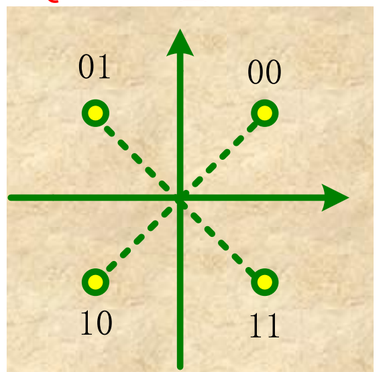
16-QAM constellation table



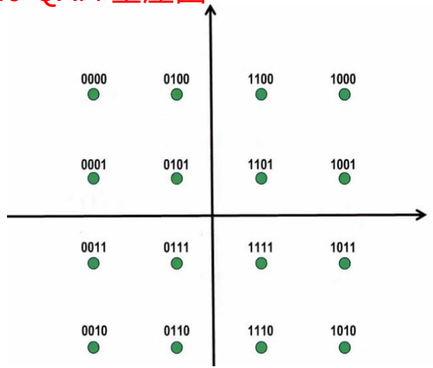
调制
(数字数据到模拟信号)

- 正交幅度调制 (QAM)
  - 相位调制和幅度调制技术
  - 如果相位有 x 次变化，幅度有 y 次变化，那么就有 x * y 次组合变化

4-QAM 星座图

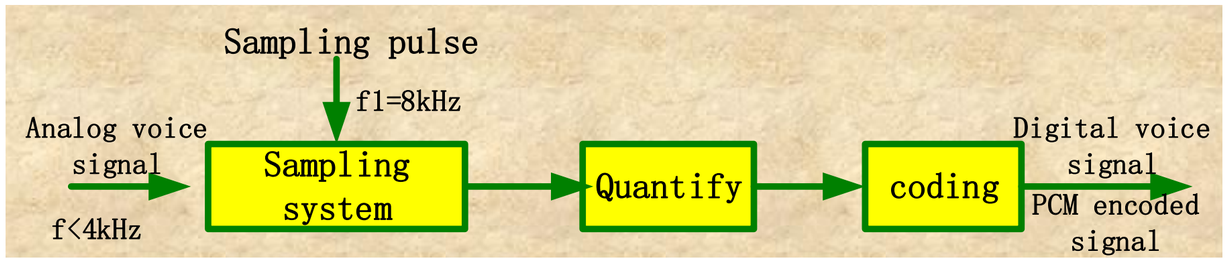


16-QAM 星座图



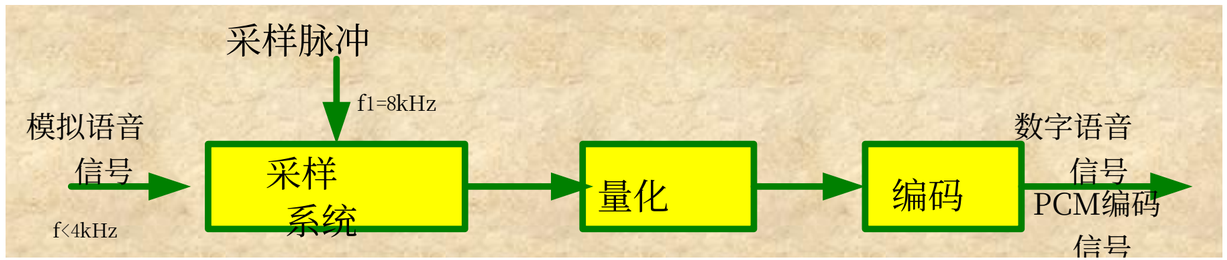
Pulse Code Modulation (PCM) (analog data to digital signals)

- Sampling Theorem
  - As long as the sampling frequency is not less than twice the maximum frequency of the signal, the original signal can be recovered from the sampling pulse without distortion
- PCM system model



脉冲编码调制（PCM）(模拟数据到数字信号)

- 采样定理
  - 只要采样频率不小于信号的最高频率的两倍，原始信号就可以从采样脉冲中无失真地恢复
- PCM系统模型



Pulse Code Modulation (PCM) (analog data to digital signals)

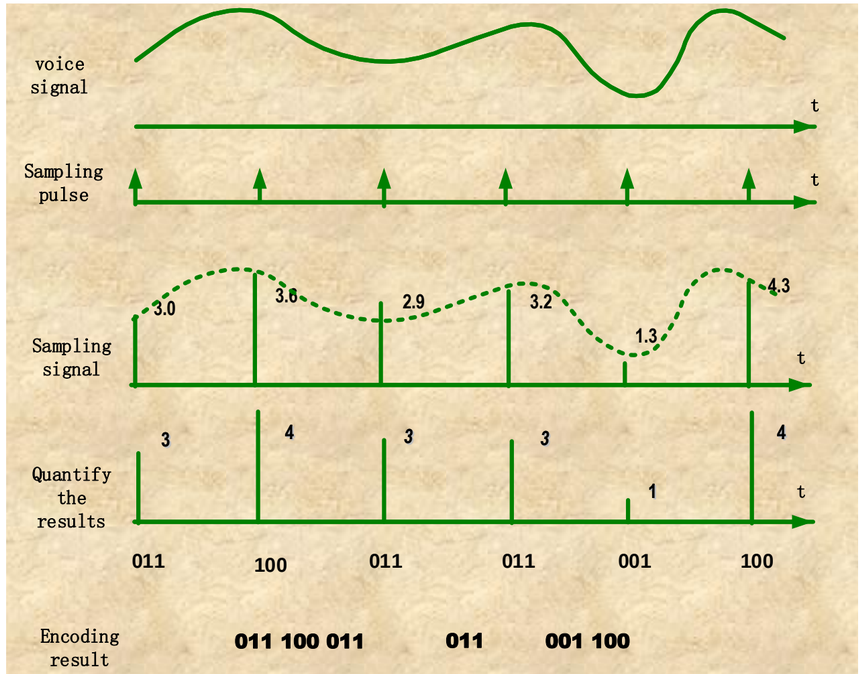
- Voice signal digitization
  - Voice frequency bandwidth f <4kHz
  - Sampling clock frequency: 8kHz (> 2 times the maximum voice frequency)
  - Sample quantization series: 256 (8bit / per sample)
  - Data rate: 8000 times / s * 8bit = 64kb / s
  - The rate of each PCM signal = 64kb / s

脉冲编码调制 (PCM) (模拟数据到数字信号)

- 语音信号数字化
  - 语音频率带宽f <4kHz
  - 采样时钟频率：8kHz (> 2 倍最大语音频率)
  - 样本量化系列：256 (8位/每样本)
  - 数据速率：8000次/秒 * 8位 = 64kb/秒
  - 每个PCM信号的速率 = 64kb / s

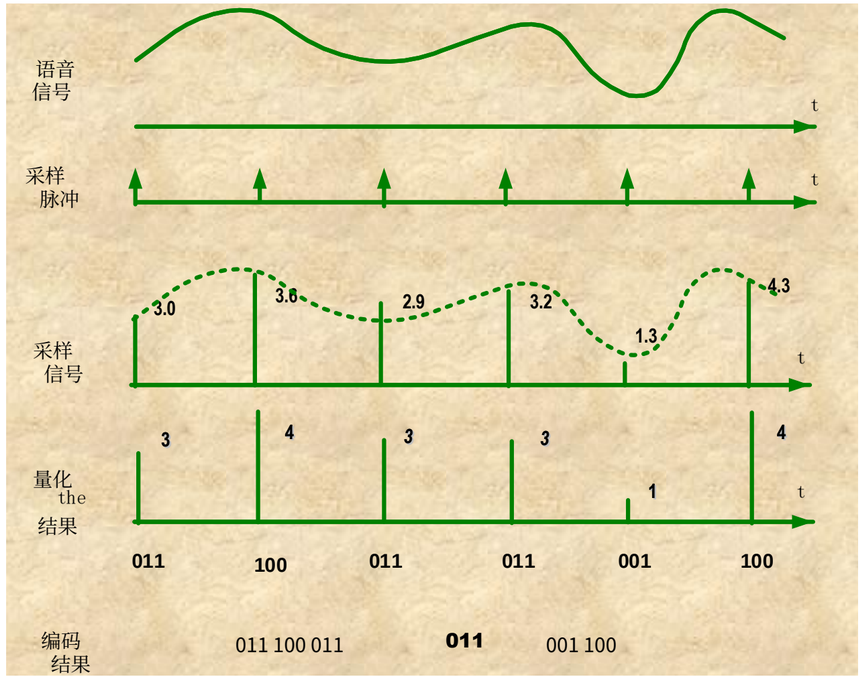
Pulse Code Modulation (PCM) (analog data to digital signals)

PCM coding process example



脉冲编码调制 (PCM) (模拟数据到数字信号)

PCM编码过程示例



2-30

Digital signal coding (digital data to digital signals)

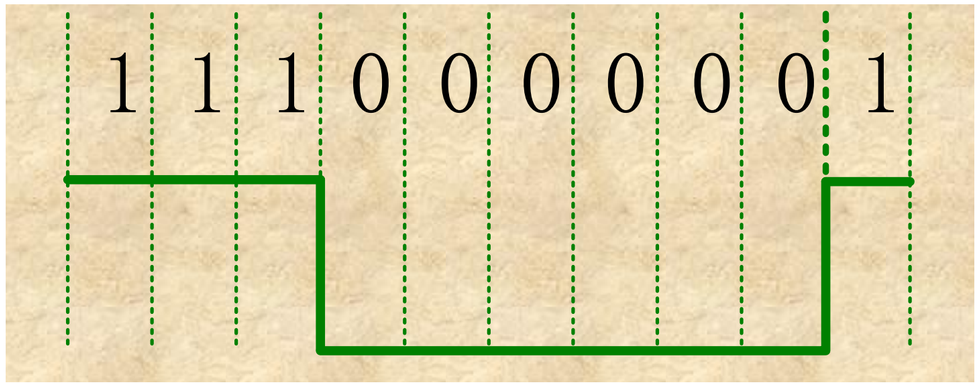
- Non-Return-to-Zero (NRZ)
- Manchester encoding
- Differential Manchester encoding
- Block encoding (4B / 5B, 8B / 10B)

数字信号编码(数字数据到数字信号)

- 非归零
- 曼彻斯特编码
- 差分曼彻斯特编码
- 块编码 (4B/5B, 8B/10B)

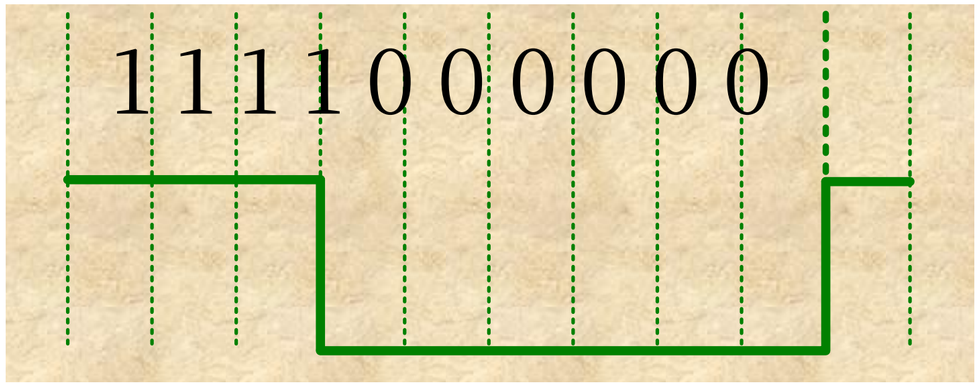
Digital signal coding (digital data to digital signals)

- Non-Return-To-Zero Level (NRZ-L) Coding



数字信号编码（数字数据到数字信号）

- 非归零电平 (NRZ-L) 编码



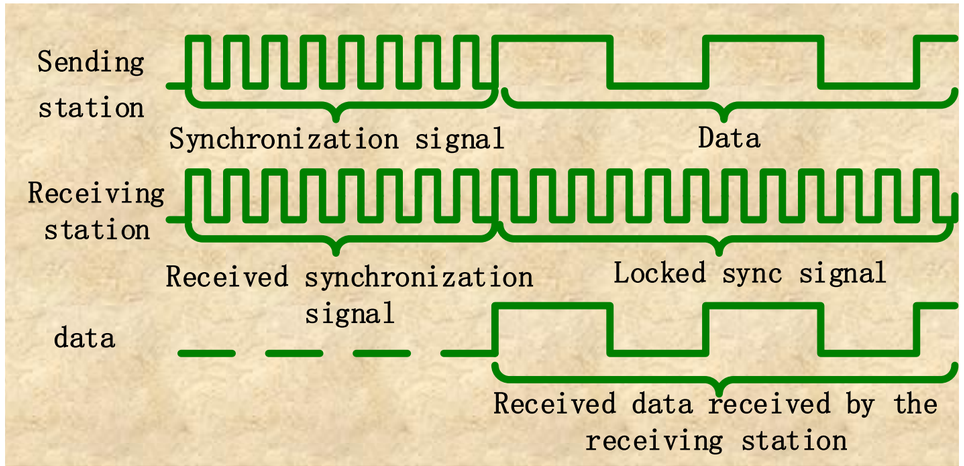
Digital signal coding (digital data to digital signals)

- Disadvantages of NRZ-L
  - Difficult to tell the end of one and the beginning of another
  - The sender and the receiver must have clock synchronization
  - If the signal "0" or "1" appears continuously, the signal DC component will accumulate
  - Prone to propagation errors

数字信号编码（数字数据到数字信号）

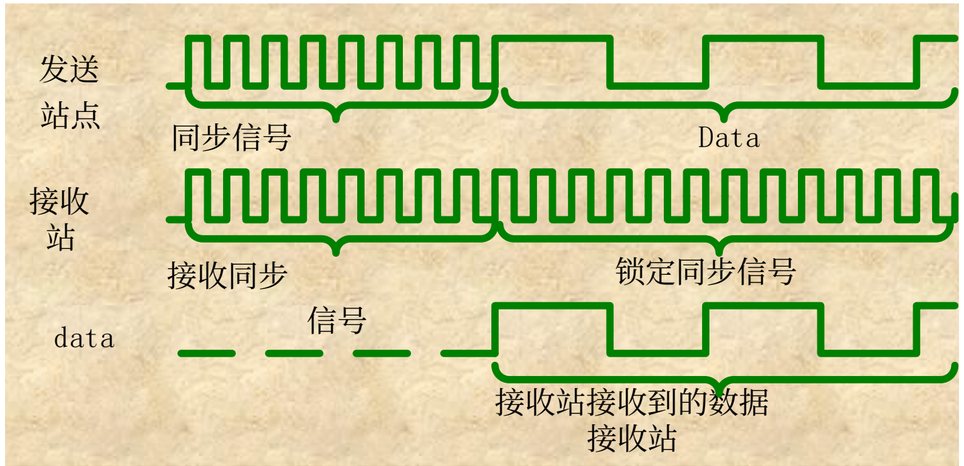
- NRZ-L 的缺点
  - 难以区分一个信号结束和另一个信号开始
  - 发送者和接收者必须实现时钟同步
  - 如果信号 "0" 或 "1" 连续出现，信号直流分量会累积
  - 易受传播错误影响

- Bit synchronization
  - The purpose is to synchronize each bit of information received by the receiver with the sender
- External synchronization



- Self-synchronization

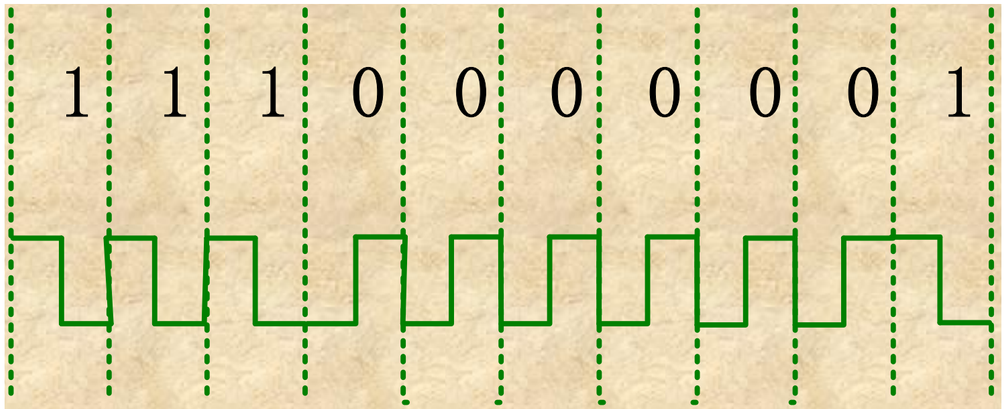
- 位同步
  - 目的是使接收者收到的每一比特信息与发送者同步
- 外部同步



- 自同步

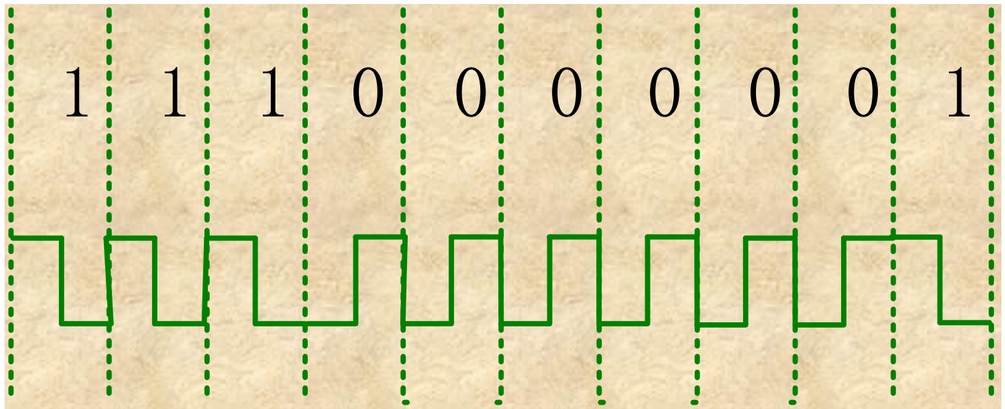
Digital signal coding (digital data to digital signals)

- Manchester Coding
  - Each symbol is divided into two equal intervals
    - "1": the previous interval is high and the next low
    - "0": the previous interval is low and the next high
  - Feature: A level hop in the middle time of each symbol



数字信号编码（数字数据到数字信号）

- 曼彻斯特编码
  - 每个符号都是 分成两个相等的间隔
    - "1": 前一个间隔为高电平，下一个为低电平
    - "0": 前一个间隔为低电平，下一个为高电平
  - 特性：每个符号中间时间存在电平跳变



Digital signal coding (digital data to digital signals)

- Advantages
  - Overcoming the disadvantages of the NRZ Coding
  - Each jump in the middle of a symbol can be used as data, and also as a clock for self-synchronization
- Disadvantages
  - Doubling the frequency bandwidth requirement of the transmission channel
  - High-frequency noise also increases – susceptible to noise interference
  - Ambiguous

数字信号编码（数字数据到数字信号）

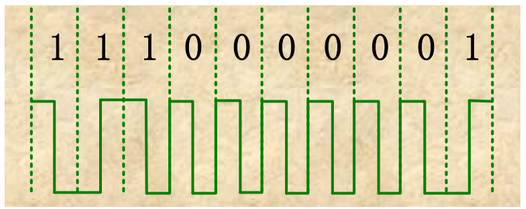
- 优点
  - 克服NRZ编码的缺点
  - 每个符号中间的跳变都可以用作 数据，同时也可以作为 自同步的时钟
- 缺点
  - 使传输信道的频率带宽需求翻倍
  - 高频噪声也会增加——更容易受到噪声干扰
  - 模糊不清

2-36

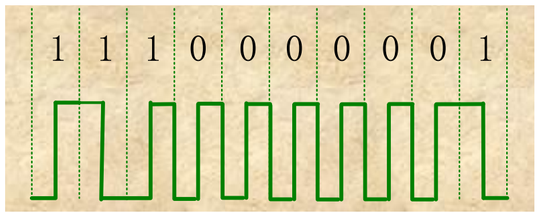
Digital signal coding (digital data to digital signals)

- Differential Manchester Coding
  - "1": the level of the first half of the symbol is the same as the level of the second half of the previous symbol
  - "0": the level of the first half of the symbol is opposite to the level of the second half of the previous symbol
- Regardless of whether the previous bit is 1 or 0

Waveform 1
Assuming the previous bit rises from low to high



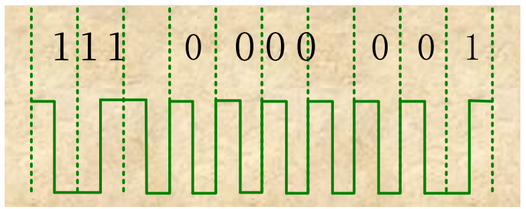
Waveform 2
Assuming the previous bit drops from high to low



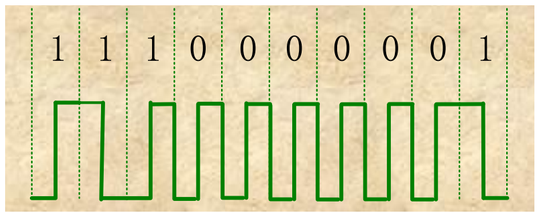
数字信号编码（数字数据到数字信号）

- 差分曼彻斯特编码
  - “1”：符号前半部分的电平与相同于前一个符号后半部分的电平
  - “0”：符号前半部分的电平与前一个符号后半部分的电平相反
- 无论前一个比特是 1 还是 0

波形1
假设前一个比特从低电平上升到高电平



波形2
假设前一个比特从高电平跳变到低电平



Digital signal coding (digital data to digital signals)

- Advantages
  - Regardless of whether the symbol is "1" or "0", there must be a level hop in the middle of each symbol => better anti–interference performance
  - Separation between bit clock and data => easy data extraction
  - Transition uniquely determined by the second half of the previous symbol => ambiguity eliminated
- Disadvantages
  - Doubling the frequency bandwidth requirement of the transmission channel
  - High-frequency noise also increases – susceptible to noise interference

数字信号编码（数字数据到数字信号）

- 优点
  - 无论符号是"1"还是"0"，每个符号中间都必须有一个电平跳变 => 更好抗干扰性能
  - 比特时钟与数据=> 分离，易于数据提取
  - 转换唯一由前一个符号的后半部分决定=> 消除歧义
- 缺点
  - 使传输信道所需的频率带宽加倍
  - 高频噪声也会增加——易受噪声干扰

Digital signal coding (digital data to digital signals)

- Exercise:

  Draw the waveform of "001101" with
  - NRZ-L
  - Manchester Coding
  - Differential Manchester Coding

数字信号编码（数字数据到数字信号）

- 练习：

  画出"001101"的波形图，使用
  - NRZ-L
  - 曼彻斯特编码
  - 差分曼彻斯特编码

Digital signal coding (digital data to digital signals)

- Block Coding
  - Advantages
    - Can improve the coding efficiency, reduce the modulation rate, and reduce the transmission line requirements
    - Some redundant bits can be added for error detection or synchronization
  - Applications
    - 4B / 5B (FDDI, 100M Fast Ethernet)
    - 8B / 10B (Gigabit Ethernet)

数字信号编码（数字数据到数字信号）

- 块编码
  - 优点
    - 可以提高编码效率，降低调制速率，并减少传输线路要求
    - 可以添加一些冗余位用于错误检测或同步
  - 应用
    - 4B/5B (FDDI, 100M快速以太网)
    - 8B/10B (千兆以太网)

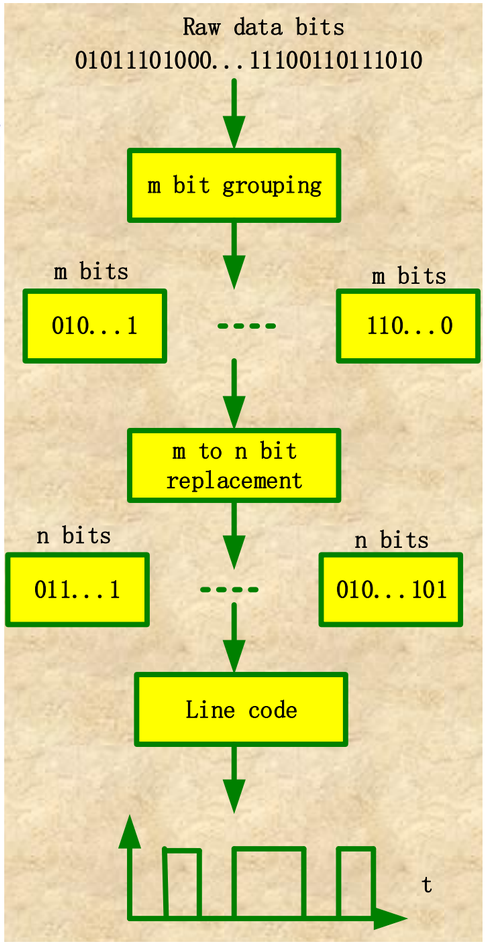
2-40

Digital signal coding (digital data to digital si

Steps (mB / nB, m < n):

- Grouping
- Substitution
  - The extra bit sequence can be used for
    - error detection
    - synchronization
    - or other controls

  or not used
- Line coding

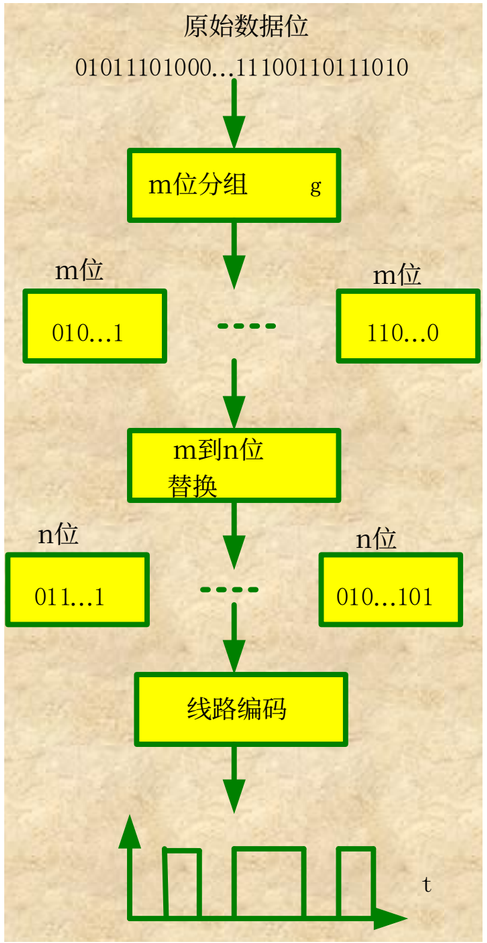


数字信号编码（数字数据到数字信

步骤 (mB / nB, m < n):

- 分组
- 替换
  - 额外的比特序列可用于
    - 错误检测
    - 同步
    - 或其他控制

  或未使用
- 线路编码



3 Basic theory of data communication

(1) Basic concepts

(2) For a specific physical communication channel, what is the maximum transmission rate? Infinite?

(3) How to transmit Bits in specific physical communication channels? Transmission speed, transmission duration, signal distortion?

(4) In order to save communication equipment and costs, how to transmit the information of multiple computers in one physical channel?

3 数据通信基本理论

(1) 基本概念

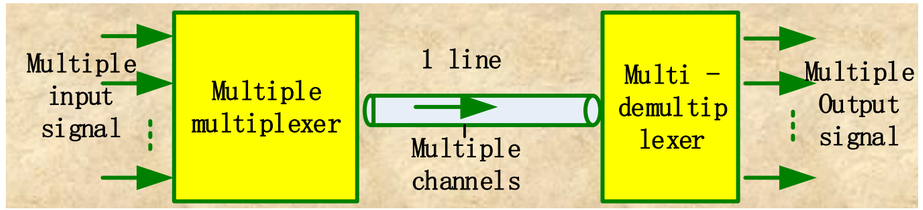
(2) 对于特定的物理通信信道，最大传输速率是多少？无限吗？

(3) 如何在特定的物理通信信道中传输比特？传输速度、传输持续时间、信号失真？

(4) 为了节省通信设备和成本，如何在一条物理信道中传输多台计算机的信息？

Multiplexing (Multi-pass)

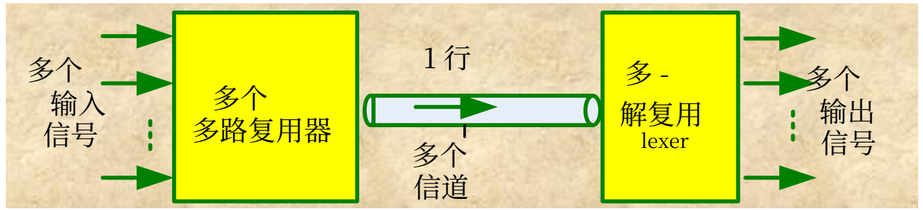
- To send multiple sources of information in one physical channel at the same time



- Commonly used techniques
  - Frequency division multiplexing (FDM)
  - Time division multiplexing (TDM)
  - Wavelength division multiplexing (WDM)
  - Code division multiplexing (CDM)

复用 (多路)

- 同时在一根物理信道中发送多个信息源

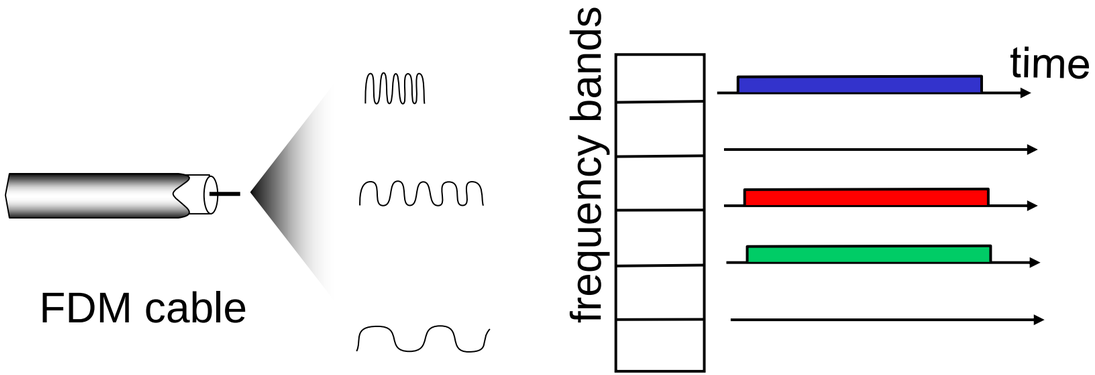


- 常用技术
  - 频分复用 (FDM)
  - 时分复用 (TDM)
  - 波分复用(WDM)
  - 码分复用 (CDM)

*Multiplexing : FDMA

FDMA: frequency division multiple access

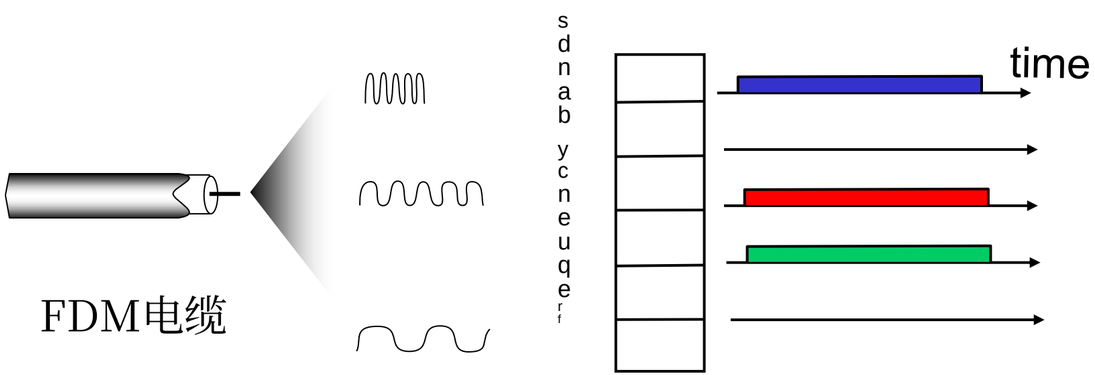
- channel spectrum divided into frequency bands
- each station assigned fixed frequency band
- unused transmission time in frequency bands go idle
- example: 6-station LAN, 1,3,4 have pkt, frequency bands 2,5,6 idle



*复用：FDMA

FDMA：频率划分多址

- 信道频谱划分为频段
- 每个站点分配固定频率带
- 频率带中的未用传输时间空闲
- 例如：6站点局域网，1、3、4有数据包，分配频率带 s 2,5,6 idle



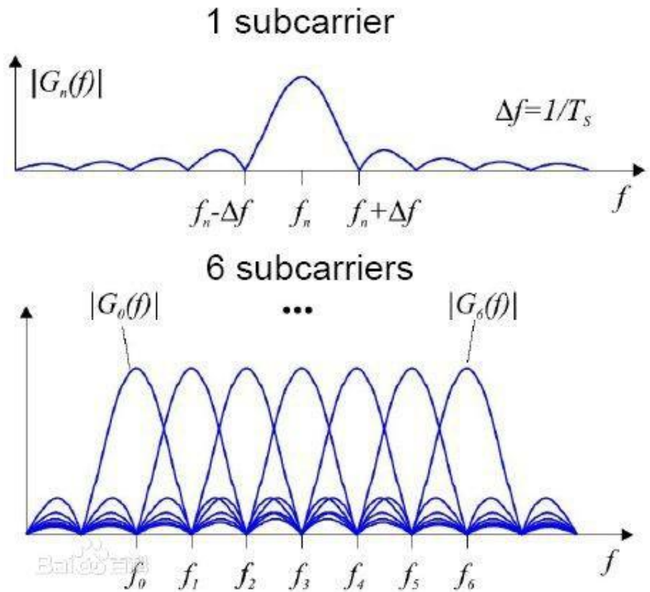
*Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology

- Basic principle:
  - Dividing the channel into a number of orthogonal sub-channels;
  - The high-speed serial data is decomposed into multiple parallel low-speed data, the use of multi-carrier FDM method to transfer, that is, the data allocated to a large number of sub-channels for transmission.
- Suitable for the high-speed data transmission through wireless channel under the presence of multipath propagation and Doppler shift.

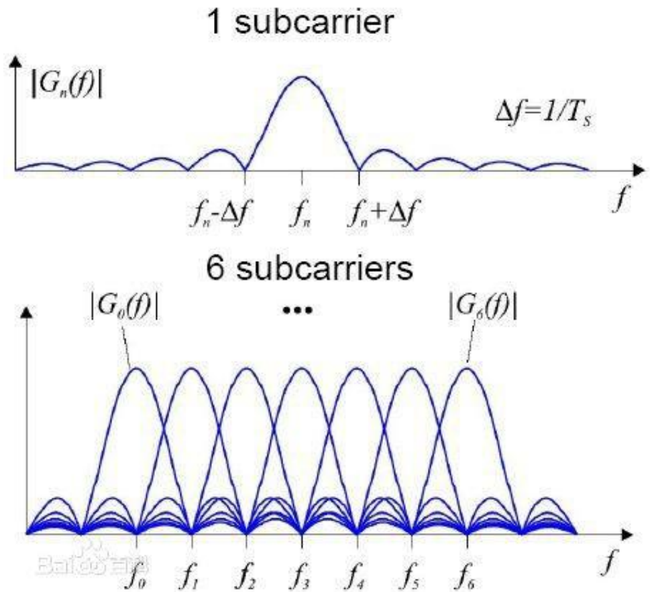
*复用：正交频分复用（OFDM）技术

- 基本原理：
  - 将信道划分为多个正交子信道；
  - 高速串行数据被分解为多个低速并行数据，采用多载波FDM方法传输，即将数据分配到大量子信道进行传输。
- 适用于在存在多径传播和多普勒频移的无线信道中进行高速数据传输。

*Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology

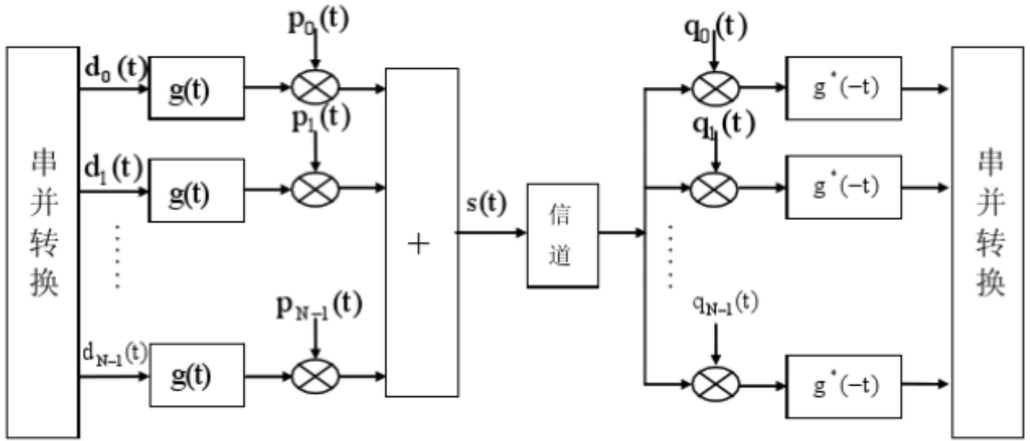


*复用：正交频分复用（OFDM）技术



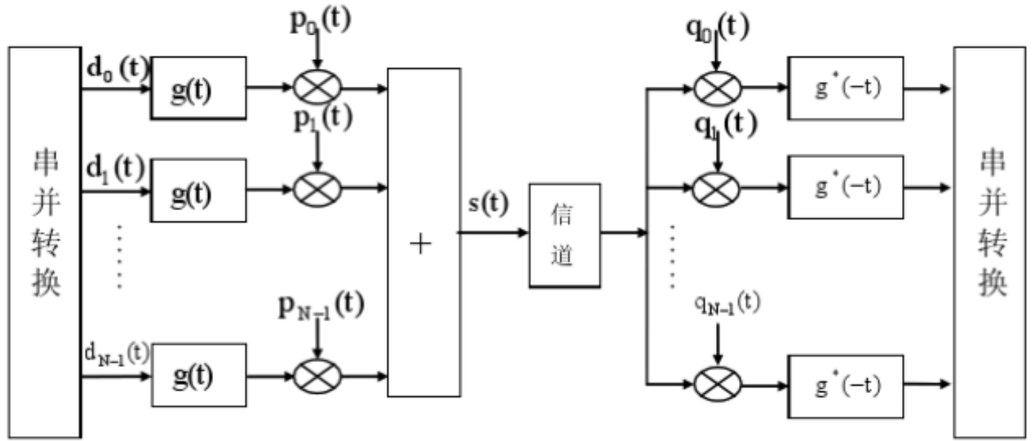
2-46

*Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology



$$\int_0^{T_s} P_m(t)q_m(t)dt = c\delta_{mn} = \begin{cases} 0 & m \neq n \\ C & m = n \end{cases}$$

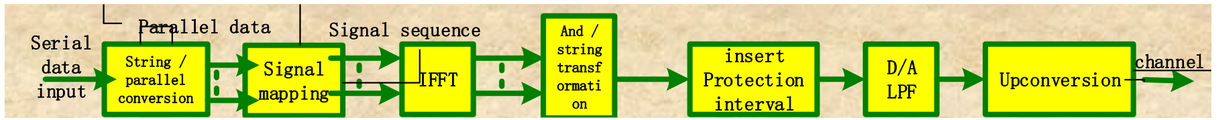
*复用：正交频分复用（OFDM）技术



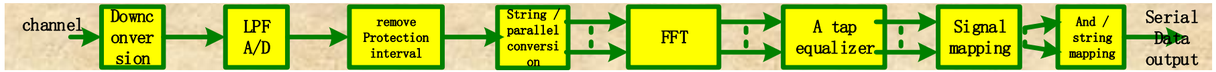
$$\int_0^{T_s} P_m(t)q_m(t)dt = c\delta_{mn} = \begin{cases} 0 & m \neq n \\ C & m = n \end{cases}$$

*Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology

- FFT based OFDM Transmit System



- FFT based OFDM Receiving System

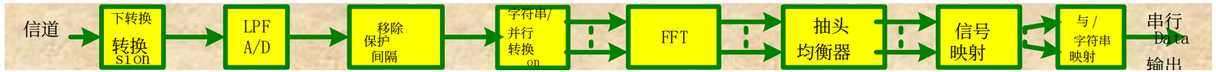


*复用： 正交频分复用（OFDM）技术

- 基于FFT的OFDM发射系统



- 基于FFT的OFDM接收系统



*Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology

- Advantages:
  - Reduce the influence of inter-symbol interference;
  - Reduce the effect of frequency selective fading;
  - Improve the band utilization, to avoid the crosstalk between sub-channels;
  - Nearly eliminate inter-symbol interference;
  - Anti-interference coding technology can be used to effectively recover the error;
  - The parallel data can be modulated and demodulated by discrete Fourier transform DFT, which reduces the complexity of system implementation.

*复用：正交频分复用（OFDM）技术

- 优点：
  - 减少符号间干扰的影响；
  - 减少频率选择性衰落的影响；
  - 提高频带利用率，避免子信道间的串扰；
  - 几乎消除符号间干扰；
  - 抗干扰编码技术可用于有效恢复错误；
  - 并行数据可以通过离散傅里叶变换DFT进行调制和解调，这降低了系统实现的复杂性。

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*Multiplexing : Orthogonal frequency division multiplexing (OFDM) technology

- Applications of OFDM technology:
  - ADSL
  - Digital Audio Broadcasting (DAB)
  - HDTV
  - Wireless LAN (WLAN)
  - Broadband radio access network
  - 3G mobile communication network and other fields.

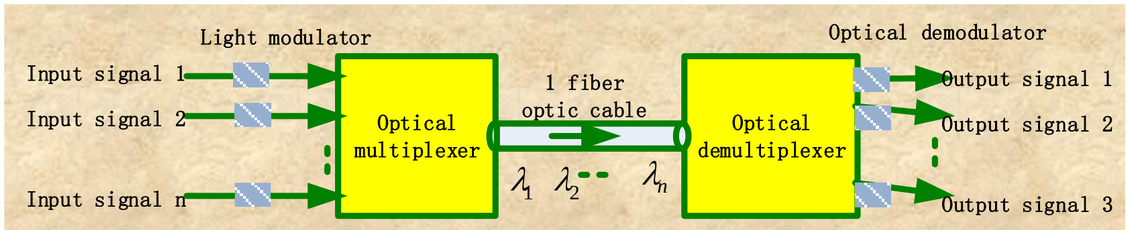
*复用：正交频分复用（OFDM）技术

- OFDM技术的应用：
  - ADSL
  - 数字音频广播 (DAB)
  - 高清电视 (HDTV)
  - 无线局域网 (WLAN)
  - 宽带无线接入网络
  - 3G移动通信网络与other字段。

2-50

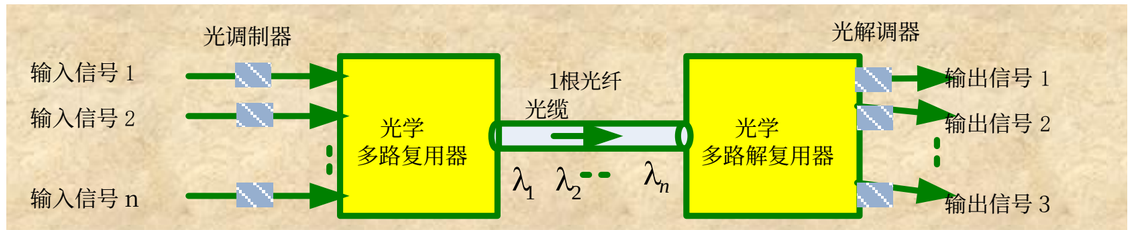
*Multiplexing : Wavelength division multiplexing (WDM)

- Transmission of multiple optical carriers with different wavelengths in a fiber at the same time
- WDM is actually a variant of FDM for fiber channel multiplexing.



*复用：波分复用（WDM）

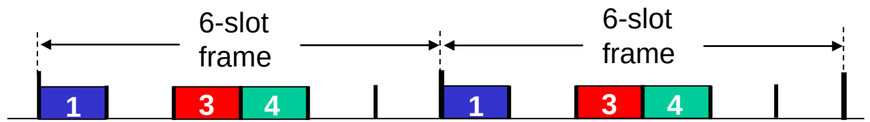
- 在同一根光纤中同时传输多个不同波长的光载波
- WDM 实际上是 FDM 在光纤信道复用中的一个变体。

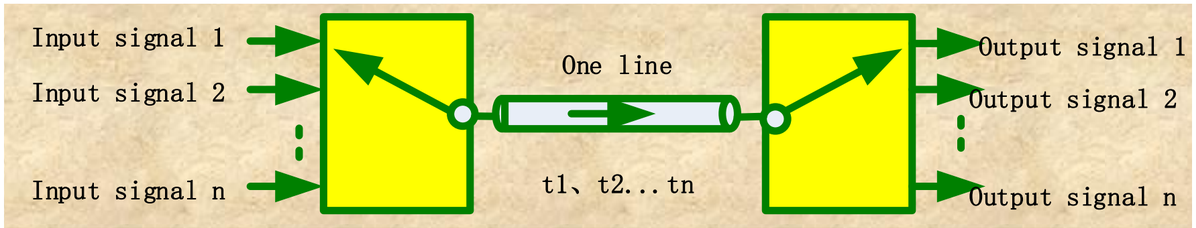


*Multiplexing : TDMA

TDMA: time division multiple access

- access to channel in "rounds"
- each station gets fixed length slot (length = pkt trans time) in each round
- unused slots go idle
- example: 6-station LAN, 1,3,4 have pkt, slots 2,5,6 idle

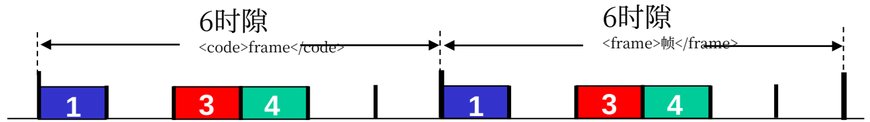


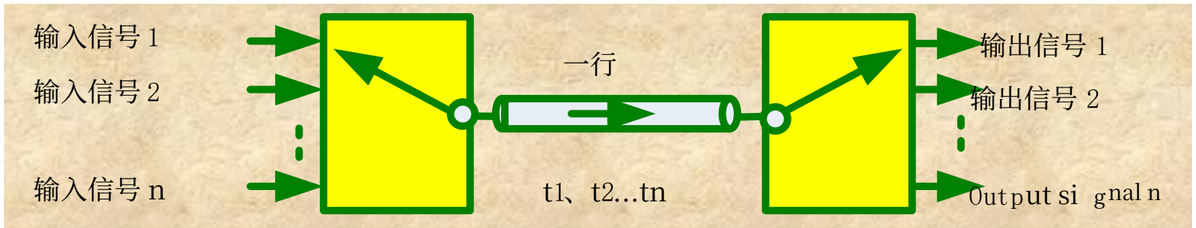


*复用 : TDMA

TDMA: 时间分片多址

- 在“轮次”中访问信道
- 每个站点在每个轮次中获得固定长度的时隙（长度 = 数据包传输时间）
- 未使用的时隙处于空闲状态
- 示例：6站LAN，1、3、4有数据包，时隙2、5、6空闲





*Multiplexing : TDMA

- Classification of TDM
  - Synchronous TDM:
    - Time slice fixed allocation, suitable for fixed rate transmission
    - The time slice of the physical channel is fixedly allocated to several users for data transmission, and each user uses the channel for data transmission when its corresponding time slice arrives.
  - Advantages: **simple implementation.**
  - Disadvantages: **there is a waste of bandwidth.**
  - **Synchronous TDM technology is suitable for fixed rate data communication system.**

*复用: TDMA

- TDM分类
  - 同步时分复用：
    - 时片固定分配，适用于固定速率传输
    - 物理信道的时片固定分配给多个用户进行数据传输，每个用户在其对应的时片到达时使用信道进行数据传输。
  - 优点：**实现简单。**
  - 缺点：**存在带宽浪费。**
  - **同步时分复用技术适用于固定速率数据通信系统。**

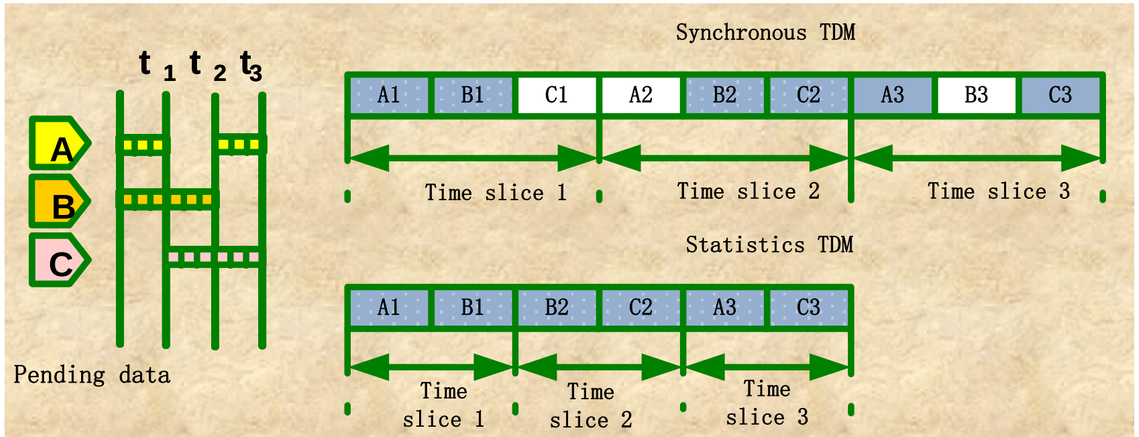
*Multiplexing : TDMA

- Asynchronous TDM:
  - The time slot of the physical channel is fixedly allocated for several, and the user does not occupy a certain time slice.
  - When a user needs to carry out data transmission, to assign a time slice;
  - If the user does not have the data transmission requirements, the system does not allocate the time slice to the user, the corresponding time slice can be allocated to other users.
- Advantages: **On-demand distribution of channel time slices, high utilization rate.**
- Disadvantages: **complex implementation.**
- **Asynchronous TDM technology is suitable for variable rate communication systems.**

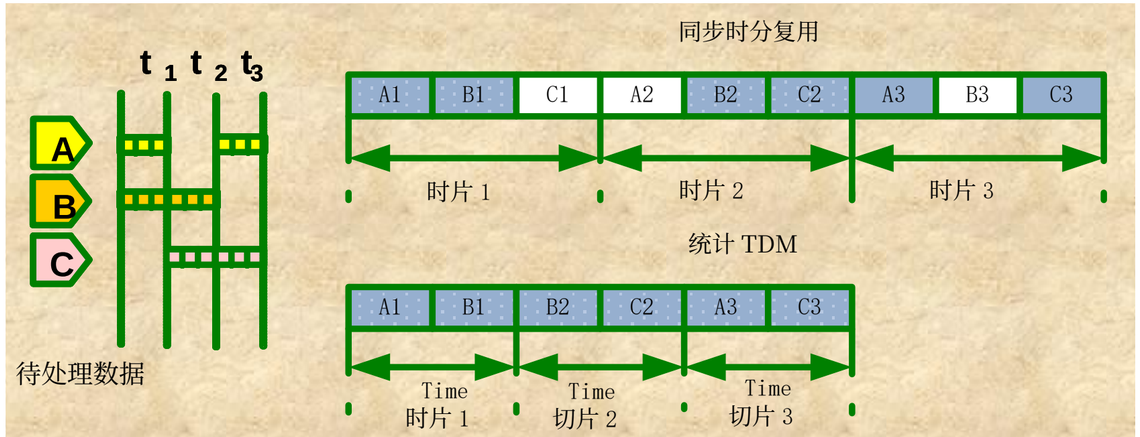
复用：TDMA

- 异步时分多址：
  - 物理信道的时隙被固定分配给多个用户，用户不占用特定的时片。
  - 当用户需要执行数据传输时，分配时片；
  - 如果用户没有数据传输需求，系统不会将时片分配给用户，相应的时片可以分配给其他用户。
- 优点：**按需分配信道时片，高利用率。**
- 缺点：**复杂实现。**
- **异步时分多址技术适用于可变速率通信系统。**

*Multiplexing : TDMA



*复用 : TDMA



*Multiplexing : Code Division Multiple Access (CDMA)

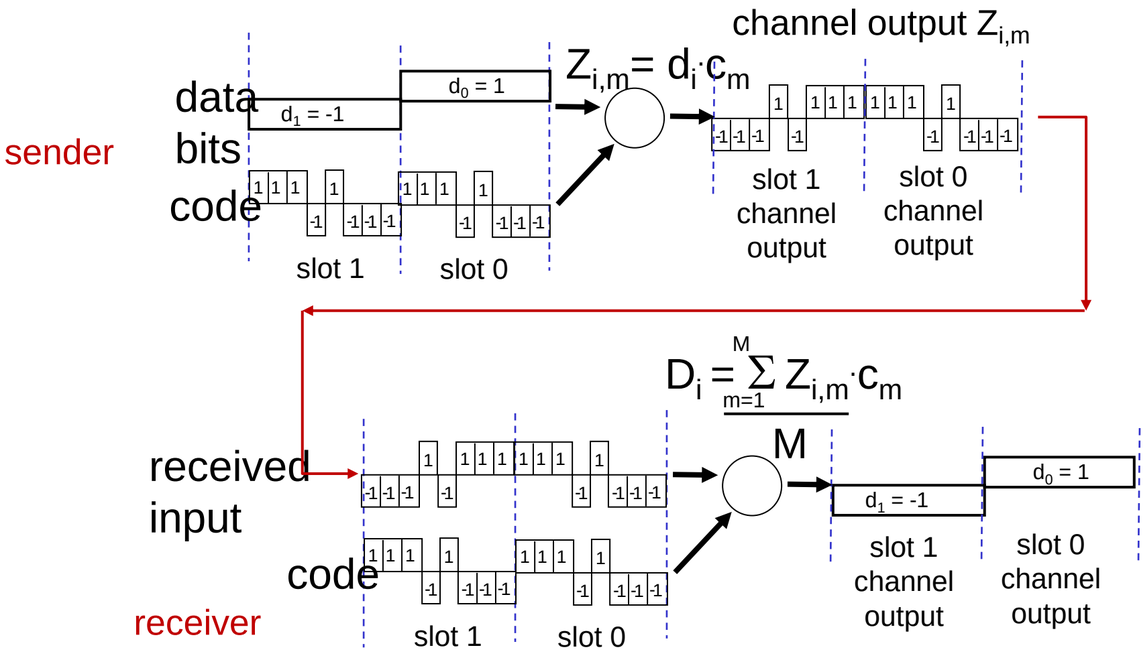
- unique "code" assigned to each user; i.e., code set partitioning
  - all users share same frequency, but each user has own "chipping" sequence (i.e., code) to encode data
  - allows multiple users to "coexist" and transmit simultaneously with minimal interference (if codes are "orthogonal")
- *encoded signal* = (original data) X (chipping sequence)
- *decoding:* inner-product of encoded signal and chipping sequence

*复用 : 码分多址 (CDMA)

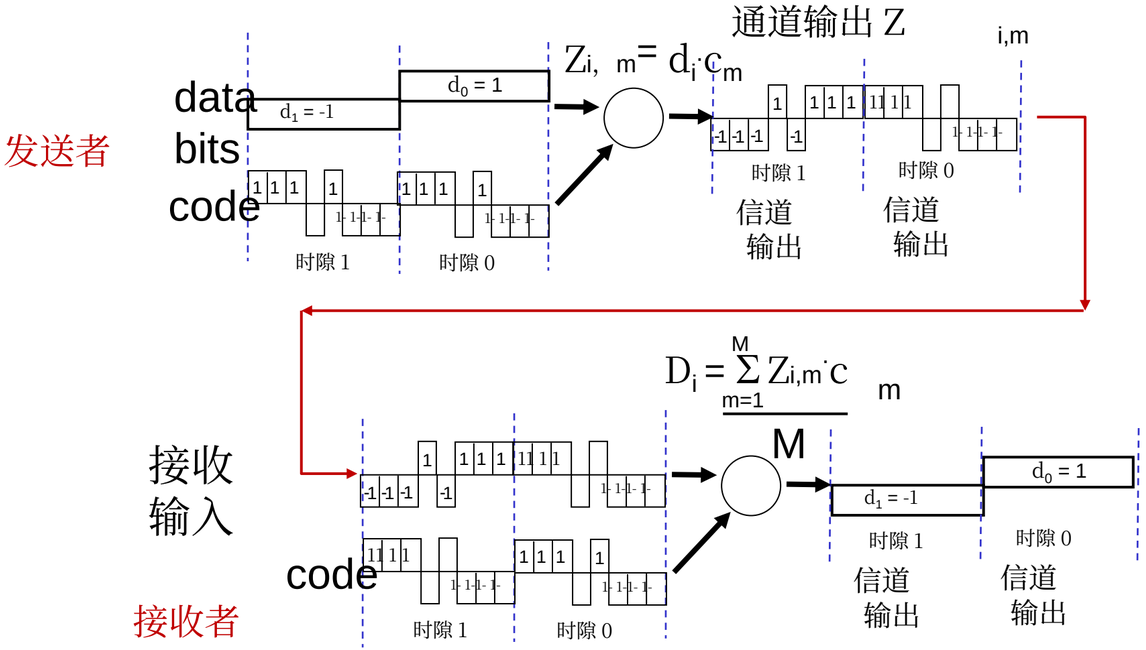
- 为每个用户分配唯一的"代码"；即，代码集划分
  - 所有用户共享相同频率，但每个用户拥有自己的"码片"序列（即，代码）来编码数据
  - 允许多个用户"共存"并同时传输，若代码是"正交"的，则干扰最小
- 编码信号 = (原始数据) X (码片序列)
- 解码：编码信号与码片序列的内积

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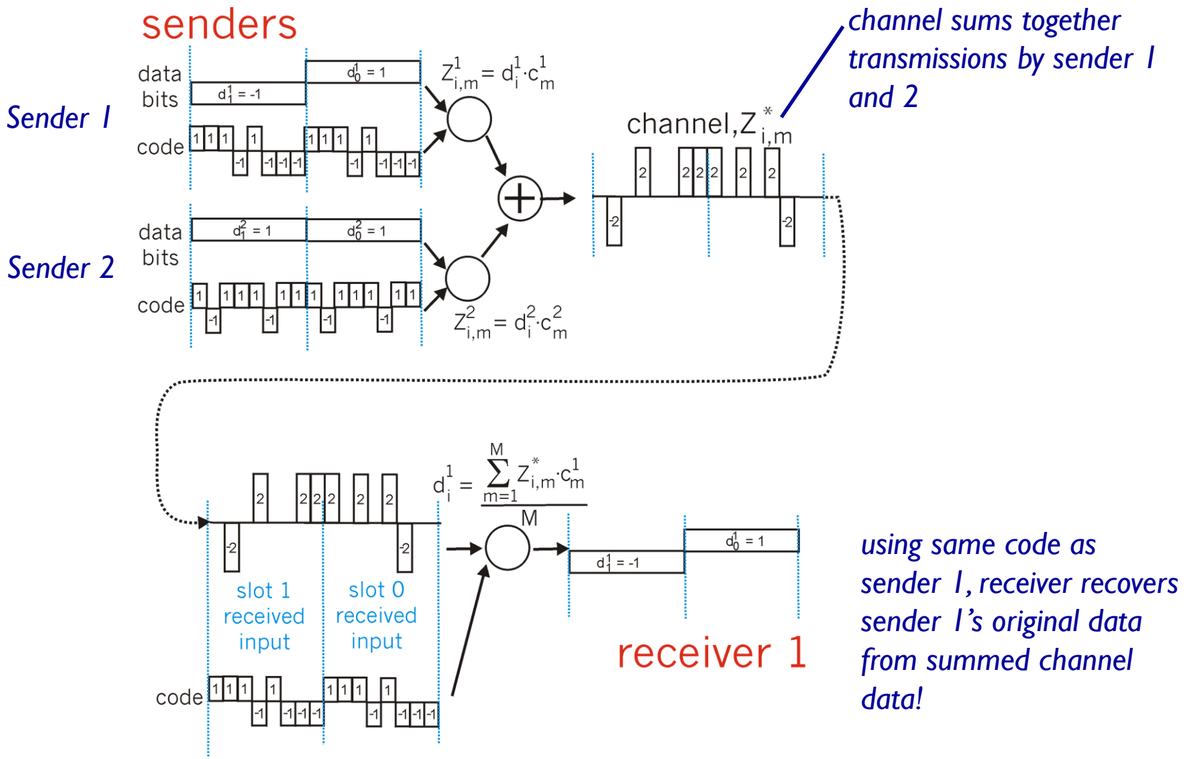
*Multiplexing : CDMA encode/decode



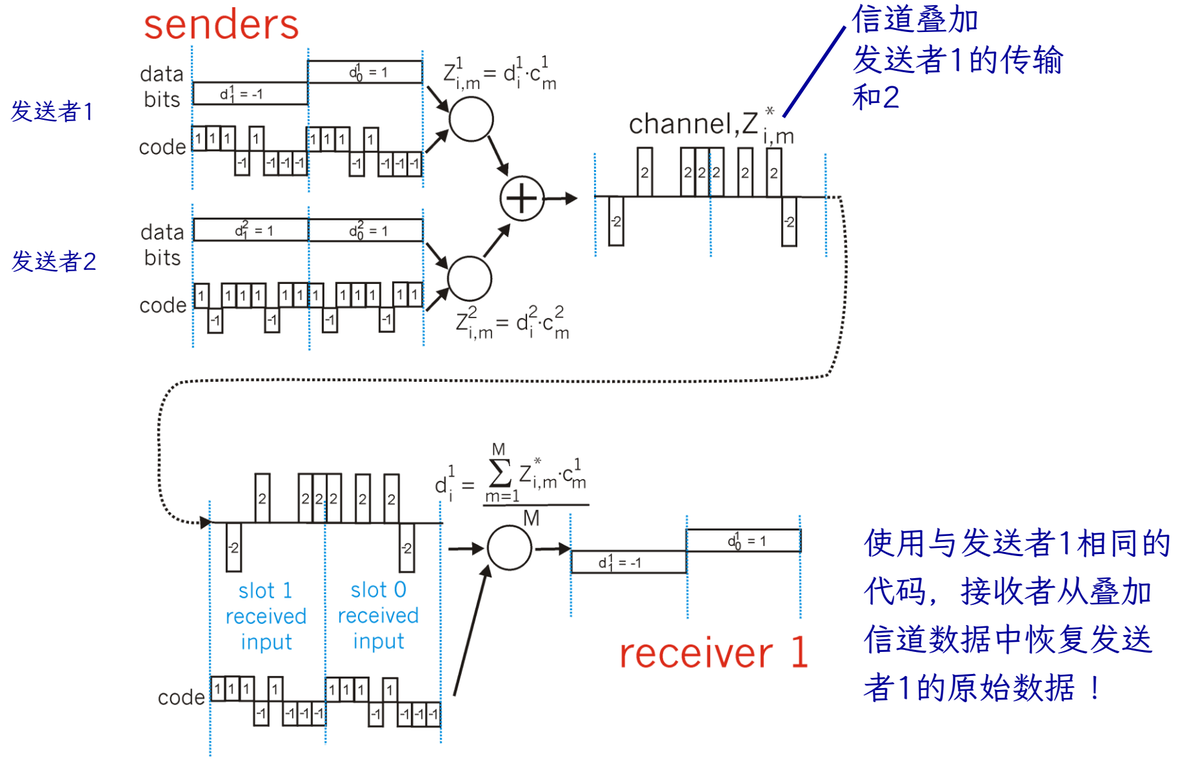
*复用 : 码分多址编码/解码



*Multiplexing : CDMA: two-sender interference



*复用 : 码分多址: 双-发信者干扰



*Comparison of commonly used media

Transmission medium	Transfer method	Rate/ Band	Transmission distance	Perform ance	Price	Application
Twisted pair	Broadband Baseband	≤1Gb/s	Anolog: 10km digtal: 500m	Better	Low	Analog / digital signal transmission
50Ω Coaxial cable	Baseband	10Mb/s	<3km	Better	Lower	Baseband digital signal
75Ω Coaxial cable	Broadband	≤450MHz	100km	Better	Lower	Analog TV, data and audio
Optical fiber	Baseband	40Gb/s	20km以上	Very good	Higher	Long - distance high - speed data transmission
Microwave	Broadband	4-6GHz	Several hundred km	Good	Medium	Remote communication
Satellite	Broadband	1-10GHz	18000 km	Very good	High	Remote communication

*常用介质比较介质

传输 介质	传输 方法	速率/ Band	传输 距离	执行 ance	价格	应用
双绞线	宽带 基带	≤1Gb/s	模拟: 10km 数字: 500m	更好	Low	模拟/数字 信号 传输
50Ω 同轴电缆	基带	10Mb/s	<3km	更好	更低	基带 数字信号
75Ω 同轴电缆	宽带	≤450兆赫	100公里	更好	更低	模拟电视，数据和音频
光纤	基带	40Gb/s	20km以上	Very good	更高	长距离 高速数据 传输
微波	宽带	4-6GHz	几个 百公里	Good	介质	远程 通信
卫星	宽带	1-10GHz	18000 km	Very good	High	远程 通信

*Commonly used physical layer interface standards

- EIA-232-E interface standard
- RS-449 interface standard
- RS-485 interface standard
- CAN interface standard
- PROFIBUS interface standard
- …….

*常用物理层接口标准

- EIA-232-E 接口标准
- RS-449 接口标准
- RS-485 接口标准
- CAN 接口标准
- PROFIBUS 接口标准
- ……..

Recap of the Physical Layer

- The function of the physical layer and the issues to be considered
- Four important characteristics of the physical layer
- The contents of the physical layer protocol
- Some Basic Concepts in Data Communication
- Nyquist's Law and Shannon's Formula
- Signal Coding Technology and Multiplexing Technology
- Commonly used transmission medias
- Commonly used physical layer interface standards

物理层回顾

- 物理层的功能及需考虑的问题

- 物理层的四个重要特性
- 物理层协议的内容
- 数据通信中的基本概念
- 奈奎斯特定律和香农公式
- 信号编码技术和复用技术

- 常用的传输介质
- 常用的物理层接口标准

Exercise-1

1.In the physical layer interface characteristics, the time sequence used to describe the completion of each function is

A. Mechanical Characteristics

B. Electrical Characteristics

C. Functional Characteristics

D. Procedural Characteristics

2. Among the following options, that do not belong to the definition scope of physical layer interface specification is

A. interface shape B. pin function

C. physical address D. signal level

练习-1

1.在物理层接口特性中，用于描述每个功能完成的时间序列是

A. 机械特性 B. 电气特性 C. 功能特性 D. 过程特性

2. 在以下选项中，不属于物理层接口规范定义范围的是

A. 接口形状 B. 引脚功能 C. 物理地址

D. 信号电平

Exercise-2

1.Under the condition of no noise, if the frequency bandwidth of a communication link is 3kHz and QAM modulation technology with four phases and four amplitudes in each phase is adopted, the maximum data transmission rate of the communication link is

A. 12kbps B. 24kbps C. 48kbps D. 96kbps

2. If the data transmission rate of a communication link is 2400bps and 4-phase modulation is adopted, the baud rate of the link is

A. 600 baud B. 1200 baud

C. 4800 baud D. 9600 baud

练习-2

1.在无噪声的条件下，如果通信链路的频率带宽为3kHz，并且采用每相位四相和四幅度的QAM调制技术，则该通信链路的最大数据传输速率是

A. 12kbps B. 24kbps C. 48kbps D. 96kbps

2. 如果通信链路的数据传输速率为2400bps，并采用四相调制，则该链路的波特率是

A. 600 波特 B. 1200 波特 C.
4800 波特 D. 9600 波特

Exercise-2

3. If the frequency bandwidth connecting R2 and R3 link is 8kHz and the signal-to-noise ratio is 30dB, and the actual data transmission rate of the link is about 50% of the maximum data transmission rate, the actual data transmission rate of the link is about

A. 8kbps B. 20kbps C. 40kbps D. 80kbps

2. Among the following factors, which will not affect the channel data transmission rate is

A. signal to noise ratio B. frequency bandwidth

C. symbol speed D. signal propagation speed

练习-2

3. 如果连接 R2 和 R3 链路的频率带宽为 8kHz，信噪比为 30dB，并且链路的实际数据传输速率约为最大数据传输速率的 50%，那么链路的实际数据传输速率约为

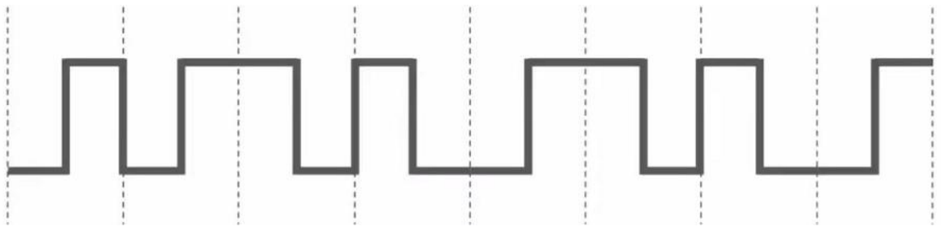
A. 8kbps B. 20kbps C. 40kbps D. 80kbps

2. 在以下因素中，哪一个不会影响信道数据传输速率

A. 信噪比 B. 频率带宽 C. 符号速率 D. 信号传播速度

Exercise-3

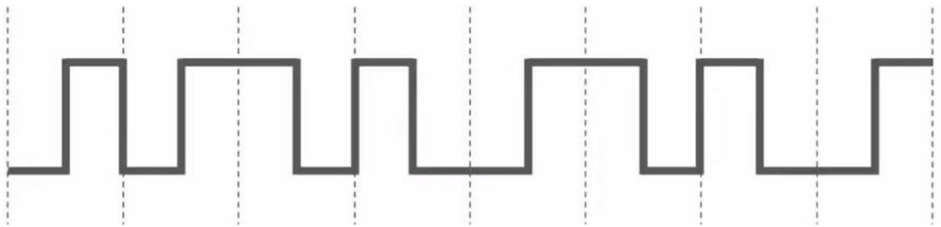
1.If the following figure shows the signal waveform received by the 10Base-T network interface card, the bit string received by the network interface card is



A. 00110110 B. 10101101 C. 01010010 D. 11000101

练习-3

1.如果下图显示了10Base-T网卡接收到的信号波形，那么网卡接收到的比特串是



A. 00110110 B. 10101101 C. 01010010 D. 11000101